Elastic Bounds of Pearson's r: Theoretical and Empirical Insights for Ordinal Variables

Cees van der Eijk   Scott Moser

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Abstract

The use of product-moment correlations to assess relationships between ordered-categorical variables is widespread and generally tolerated in spite of the non-metric character of such variables. Potential problems of this practice have been identified in the literature. This article focusses on one of these, namely that correlation coefficients are poorly comparable across pairs of ordered-categorical variables because their ranges are constrained in non-uniform ways. This not only affects the validity of correlation coefficients for descriptive purposes, but also as bases for subsequent analyses, as in, e.g., principal component analysis, factor analysis, and structural equation modelling. This article focusses on this problem of 'constrained' correlations. It explains the problem and explores conditions under which it is most severe. It provides illustrations based on both fabricated and actual empirical data, the latter from a mass-survey that ubiquitously contains many ordered-categorical variables that are commonly analysed with product-moment correlations. The article is accompanied by a software tool in R for analysing the attainable upper- and lower bounds of correlations in actual data.

Pearson's correlation coefficient r, when applied to ordinal data, is subject to elasticity due to constraints from marginal distributions. This paper develops theoretical bounds for r, based on Boole–Fréchet–Hoeffding inequalities and rearrangement inequalities, and derives analytic expressions for their limits. We investigate when symmetry in the bounds holds or breaks, and explore practical implications for correlation-based methods like PCA and SEM. An accompanying R package allows users to compute bounds and perform hypothesis tests given fixed marginals.

1 Introduction

In this article we elaborate why correlations involving ordered-categorical variables are generally 'constrained', which means that their attainable maximum and minimum values are rarely +1 and -1, as is often thought, but values that have to be calculated for each pair of variables. We assess the magnitude of these constraints in fictitious as well as empirical data and analyse under which conditions they are most severe. The article is accompanied by a software tool in R: that allows applied researchers to easily assess the extent of the problem in their own data.

The use of product-moment correlations (also known as Pearson correlations) to assess linear relationships between ordered-categorical variables –henceforth referred to as orcats– is widespread and widely condoned in empirical studies, in spite of such variables not being metric while the product-moment correlation assumes that they are. We do not focus on this problem from an obsessive desire for methodological purity, were it only because many statistical procedures are quite robust for violations of their assumptions [fn: examples?]. Instead, we draw attention to problematic consequences of this violation of assumptions that, as far as we know, have been rarely recognised in the applied literature although they are well-known in the theoretical statistical literature. This is the problem that correlation coefficients involving orcats are constrained in their range which makes them mutually incomparable, leading to problems in hypothesis testing and in analytical procedures that use correlations as their input.

Despite long-standing cautions about applying Pearson's correlation coefficient r to ordinal variables (Gaito, 1980; Labovitz, 1970), it remains routine in survey-based social science. In political science and sociology alike, researchers frequently correlate 4–, 5–, or 7–point items as if they were interval-scaled, usually without examining the implications of the marginals. Examples include correlations among attitude items in large-scale survey batteries (Alwin & Krosnick, 1991), ideology and issue positions on mixed-resolution scales (Brewer & Gross, 2005; Kinder & Kam, 2009), and summary indices that are subsequently modeled using correlation-based techniques. Reviews document that such practices are pervasive in applied survey analysis (Kreuter & Valliant, 2007). Methodological discussions sometimes defend pragmatic use of parametric tools with ordinal data (Jacoby, 1999), but even optimistic views do not address the *feasible range* of r imposed by fixed marginals—an omission with direct consequences for substantive interpretation.

Alternatives to Pearson's r—notably Spearman's ρ, Kendall's τ, and the polychoric correlation—offer different trade-offs (Kendall, 1938; Olsson, 1979; Spearman, 1904). Rank-based measures ignore the marginals; polychorics assume latent bivariate normality. In contrast, we pursue bounds on r that *explicitly* depend on the observed marginals, drawing on rearrangement inequalities and the Fréchet–Hoeffding framework (Fréchet, 1951; Hoeffding, 1940; Rüschendorf, 1981). This perspective clarifies when the attainable range of r is narrow, when symmetry fails ($r_{\min} \neq -r_{\max}$), and when the lower bound can be strictly positive. It also explains why correlation-based models with ordinal indicators (EFA/CFA/SEM) can be distorted when Pearson correlations are used (Flora & Curran, 2004; Green et al., 1997). Our contribution situates empirical practice within these theoretical limits and provides tools to compute and visualize the bounds given fixed marginals.

This article proceeds as follows. Section 2 introduces orcats, explains why Pearson's r is used despite violating the interval-level assumption, and formally defines the notation and key concepts — including existing alternatives and the coupling framework that underlies our bounds. Section 3 derives the theoretical results: the main proposition establishing $r_{\min}$ and $r_{\max}$ via comonotonic and anti-comonotonic coupling, the conditions under which $|r_{\min}| = r_{\max}$ (symmetry) and $|r_{\min}| > r_{\max}$ (asymmetry), and results on the signs of extreme correlations. Section 4 introduces metrics of constraint — scalar summaries of how severely marginal distributions restrict the attainable range of r. Section 5 presents Monte Carlo validation across orcat pairs with 4 to 11 categories, characterising how marginal shape drives constraint. Section 6 illustrates the findings using 7,503 variable pairs from the British Election Study 2019. Section 7 draws out applied consequences for effect-size interpretation, cross-study comparability, hypothesis testing, rescaling, and the use of correlation matrices in PCA, FA, and SEM. The article concludes with a discussion and directions for future research.

2 Background: Orcats, Pearson's r, and Fixed Marginals

2.1 Ordinal-Categorical Variables: Definition and Common Use

Orcats are ordinal variables with a relatively small number of categories (or ranks) compared to the number of cases, thus resulting in partial orderings of cases. Although they can be found in all field of quantitative inquiry they are ubiquitous in data from standardised social science surveys. Likert items and rating scales [describe in fn Likert items and rating scales, add that there are yet other formats such as semantic-differential scales that also yield orcats] are among the most favoured ways to elicit responses in such surveys. These survey-question formats have most often between 3 and 11 ordinal categories. The numerical values assigned to the categories of orcats are most often consecutive integers, starting with either 0 or 1. [fn: explain that these values may be included in the formulation of a survey question, which is immaterial to our discussion here. The values assigned to the categories (and to the responses) reflect, in combination with the analysis procedure the analyst's interpretation of their meaning.] . Such an assignment of values is as good as, and indeed analytically equivalent to, any other set of values that reflects the analyst's understanding of the ordinal relations between the categories, i.e., any monotonic transformation. However, objections are often raised when orcats [fn: add references] are analysed with procedures requiring metric measurement. That requirement implies that the differences between values can validly be compared so that, e.g., the substantive difference between categories 1 and 2 is equal to that between categories 3 and 4, and half as large as the difference between categories 2 and 4, etc. Any implied interval-level interpretation of ordinal categorical variables requires therefore a compelling theoretical or empirical justification of the numerical values assigned to the categories rather than just the analyst's fiat. In principle this is quite possible, for example by calibrating values of categories against psychophysical measurements (cf., Lodge; Tillie; others), or against the results of psychometric/linguistic analysis of verbal response labels (cf Sönning 2024). However, such procedures are rarely integrated in applied survey research because of their high cost. In the absence of compelling justifications for values assigned to categories orcats can only be considered to provide ordinal information, not metric information. In spite of this we find that they are very commonly analysed with statistical tools that require metric measurement of variables such as product-moment correlations or procedures that build upon such correlations. We find this clearly exhibited in published articles in high-ranking academic journals [fn: explain how we did this] and in highly regarded textbooks where examples are given of the use of correlations between orcats as input for procedures such as PCA, FA or SEM [fn: examples: Kline; Byrne; xxx]. This practice is so established that it is occasionally rationalised by referring to orcats as being 'quasi-interval level' measures [fn: reference]. One can, however, also find more explicit justifications for this practice, which emphasise the robustness of correlation coefficients for violations of their measurement level assumptions. Labovitz (1967, 1970) in particular has compared correlations between orcats scored according to equal-distances with correlations when the variables were scored with a monotonic random procedure (which only assumes ordinality between categories). He reports that the differences are negligible, and suggests that even if the theoretical objection against unfounded interval level interpretations of orcats would be justified, that objection would nevertheless have such minor practical consequences as to make it irrelevant in most situations. Not surprisingly, Labovitz' work has in turn be criticised itself on various grounds, most particularly of (inadvertently) mainly covering situations where this robustness holds, and overlooking other situations (cf O'Brien 1979; xxx). At this place we will not focus on discussions about the equal-distance scoring practice. Our main reason for referring to them is to demonstrate the existence of lively debates in the applied literature about potential issues when using orcats in procedures requiring metric data. The existence of these debates makes it remarkable that another important issue has –to the best of our knowledge– remained mainly overlooked in the applied literature while it has been well-known in the theoretical statistical literature. This is the problem that correlations between orcats are generally 'constrained', which means that they do not range between -1 and +1, but between minima and maxima that have to be calculated separately for each pair of variables. The nature of this issue is the subject of the next section.

2.2 Correlation Bounds for Orcats

The values that product-moment correlations between ordinal-categorical variables (orcats) can attain are generally constrained by minimum and maximum bounds of attainable values that are not -1 and +1. [fn: The maximum attainable value is +1 only when the univariate distributions of the variables are identical. But that does not imply the minimum attainable value then to be -1, as that also requires further conditions that the univariate distributions ,must meet (see below, in the discussion of Table 2). The minimum attainable value of a correlation is -1 when the univariate distributions are each other's inverse, but again, that does not imply that the maximum is +1, as that requires additional conditions of the univariate distributions.]

Because the bounds of correlations between orcats are not -1 and +1 it is even more difficult than commonly thought in applied research to interpret the strength of correlations. The well-known and often referred to criteria formulated by Cohen (1988: 77-81) for correlations in the behavioural sciences [fn: Cohen suggested, with appropriate caution, the following guidelines: $0.10 < r < 0.29$ weak; $0.30 < r < 0.49$ moderate; $0.50 < r < 1.00$ strong. Hemphill (2003) suggests on the basis of a review of several meta-analyses in psychological research that the cut-off of 0.50 for considering a correlation to be 'strong' may be too high in many common instances.] are explicitly based on the notion that the minimum and maximum attainable values for r are -1 and +1. But that is generally not the case and, moreover, the actual upper- and lower-bounds of r are not the same across pairs of variables, but specific to each. This state of affairs thus generates two interconnected problems when correlating orcats. First, the absence of clarity of the lower- and upper-bounds of r makes qualitative interpretations of the strength of correlations elusive. Second, because the bounds of attainable values of r are specific for each pair of variables each correlation is expressed in terms of a different, but unknown reference interval. Such correlations are therefore not comparable. To better grasp these problems, it is necessary to determine the upper- and lower bounds of correlations, given the univariate distributions of the variables involved.

Statistical scholarship has since long recognized that, in contrast to what is ofen thought, the upper- and lower-bounds of correlations between orcats are generally not the reference interval [-1, +1]. [insert references, including Shi&Huang 1992; xxxx]. Determining the actual bounds given the marginal distributions of the two variables can be done by using Whitt's (1976) rearrangement theorem and Hardy–Littlewood–Pólya's (1952) rearrangement inequality. For obtaining the maximum attainable value of a correlation the values of X and Y have each to be sorted independently in either ascending or descending order and then paired (comonotonic arrangement). For the minimum attainable value of the correlation (given the marginals of the two variables) the anti-comonotonic arrangement is required (sorting one of the variables in ascending and the other in descending order). A formalized elaboration is presented in Appendix 1. Moreover, the algorithm for calculating rmin and rmax has been programmed in R and is [or: will be] freely available for researchers to assess the minimum and maximum attainable values of r for pairs of variables in their own data. Details about this routine are described in Appendix 2.

A series of illustrative cases is presented in Table 2 (as in Table 1 for n=12 and two orcats with 4 categories each). Although the distributions of the variables given in these examples is of no particular importance, and could have been replaced by innumerable other examples, they are nevertheless helpful to arrive at some general observations about rmin and rmax.:

- The values of rmin and rmax are not constant across all examples, and not even across correlations involving a specific variable (see examples (a) and (b) which involve the same X-variable). The variation in the values of the bounds of r is across pairs of variables. In other words, the bounds of r are specific for each pair.
- rmin max can be -1, and rmax can be 1, but neither of these values implies the other (as seen in examples (b) and (c). The combination of rmin=-1 and rmax=1 is possible, but dependent on specific characteristics of the two marginal distributions involved (see examples (d) and (f). Both variables must not only have identical distributions,

but these must also be symmetrical.

- rmin and rmax are not necessarily symmetric around 0. Such symmetry may exist, but requires specific characteristics of the pair of marginal distributions, namely that both are symmetric (see examples (e), (g) and (h).

- The extent to which the reference interval [-1, 1] is covered by the interval [rmin, rmax] varies across pairs of variables.
- The values of rmin and rmax depend not only on the marginal distributions of the two variables, but also on the values assigned to their categories. Unless specified differently, we follow in this article the common practice of assigning consecutive integer values to the ordered categories. But any monotonic transformation of values is equally adequate to represent the ordinal character of variables. Examples (i) and (j) involve the same pair of variables, but in (i) their categories have been assigned values 1. 2, 3 and 4, while in (j) the assigned values were 1, 2.25, 2.75 and 4. The consequences of this change are visible in the values of rmin and rmax. This observation implies that value assignment to categories of orcats (and the justification thereof) remains an issue of importance when using Pearson correlations on orcats. It is a different problem, however, than that of constrainment of correlations on which we focus here.

2.3 Consequences for Applied Research

The constrained range of r_{XY} has two direct consequences for how correlations between orcats are interpreted and compared in applied research.

First, qualitative benchmarks for correlation strength — most famously, Cohen et al. (2002)'s suggestions of $r \approx 0.10$ (weak), 0.30 (moderate), and 0.50 (strong) — presuppose that the attainable range is $[-1, +1]$. When the actual range for a given pair of variables is, say, $[-0.65, +0.72]$, an observed $r = 0.40$ lies considerably closer to the maximum than it would appear under Cohen's scheme. The benchmark is inapplicable, and applied conclusions about effect strength are distorted in a direction and by a magnitude that varies unpredictably across variable pairs.

Second, and more fundamentally, every pair of orcats has its own unique attainable interval $[r_{\min}, r_{\max}]$, which means that correlations from different variable pairs are expressed on incommensurable scales. To see why this matters, consider two pairs (A, B) and (B, C) with observed correlations $r_{AB} = 0.30$ and $r_{BC} = 0.45$. If $r_{\max}^{AB} = 0.40$ while $r_{\max}^{BC} = 0.85$, then (A, B) is near its theoretical maximum while (B, C) is only moderately constrained. The conclusion that B is more strongly associated with C than with A may well be reversed once the different attainable scales are accounted for. Without examining the bounds, comparisons of correlations across variable pairs are uninformative and potentially misleading.

These two problems motivate the development, in subsequent sections, of constraint metrics that summarise the severity of marginal restriction, and of rescaling strategies that place observed correlations in their proper bounded context.

2.4 Background and Definitions

Notation. Let X take values $x_1 < x_2 < \cdots < x_{K_X}$ with marginal probability distribution $p_X = (p_{X,1}, \ldots, p_{X,K_X})$, and let Y take values $y_1 < y_2 < \cdots < y_{K_Y}$ with marginal probability distribution $p_Y = (p_{Y,1}, \ldots, p_{Y,K_Y})$, where $\sum_i p_{X,i} = \sum_j p_{Y,j} = 1$. In survey research, categories are typically scored with consecutive integers beginning at 0 or 1. Any monotone transformation of these scores is equally valid under ordinality, but Pearson's r is sensitive to the choice of scoring — unlike Spearman's ρ, which is invariant to monotone transformations, Pearson's r is only invariant under affine (linear) recoding.

The Pearson product-moment correlation is:

$$r_{XY} = \frac{E[XY] - E[X]E[Y]}{\sigma_X \sigma_Y},$$

where $E[\cdot]$ denotes expectation under the joint distribution of (X, Y), and σ_X, σ_Y are the marginal standard deviations. Since $E[X]$, $E[Y]$, σ_X, and σ_Y depend only on the marginal distributions, maximising or minimising r_{XY} over all joint distributions consistent with fixed marginals reduces to maximising or minimising $E[XY]$.

Existing alternatives. Several measures have been proposed to address the limitations of Pearson's r with ordinal data. Spearman's ρ (Spearman, 1904) replaces raw values with ranks and is thus insensitive to the specific marginal distributions; it discards information about inter-category spacing. Polychoric correlation (Olsson, 1979) assumes that X and Y are discretisations of a latent bivariate normal distribution and estimates the latent correlation under that assumption. Polyserial correlation (Olsson, 1979) extends this to the case where one variable is continuous. All three alternatives abstract away from the observed marginals. In contrast, our approach retains the observed marginals and uses them to characterise the feasible range of Pearson's r — a quantity that is directly relevant to applied researchers who report and compare Pearson correlations.

Coupling and rearrangement. The key insight is that fixing p_X and p_Y places hard constraints on the set of admissible joint distributions, and therefore on the attainable values of r_{XY}. The Fréchet–Hoeffding bounds (Fréchet, 1951; Hoeffding, 1940) characterise the extremal joint distributions with given marginals. In the discrete setting, the constructive tool is the rearrangement inequality of Hardy et al. (1952): for two finite sequences, their sum of products is maximised when both sequences are sorted in the same order (comonotonic arrangement) and minimised when sorted in opposite orders (anti-comonotonic arrangement). These constructions yield $r_{\max}$ and $r_{\min}$ respectively, and are formalised in Section 3.

3 Theoretical Results

The problem of extremal correlations under fixed marginals connects directly to classical results in probability inequalities. Fréchet's work on admissible joint distributions ((Fréchet, 1951)) and Hoeffding's development of covariance bounds ((Hoeffding, 1940)) laid the groundwork. Later refinements, such as Rüschendorf's demonstration of sharpness of Fréchet bounds ((Rüschendorf, 1981)), clarified when comonotonic and countermonotonic couplings achieve extremal values. The rearrangement inequality, and its discrete extensions, ensure that matching high values of X with high values of Y maximizes $\mathbb{E}[XY]$, while opposite pairings minimize it. This perspective is elaborated in comparison and coupling frameworks (Müller & Stoyan, 2002; Thorisson, 2000). These results provide the theoretical backbone for our derivations of $r_{\min}$ and $r_{\max}$.

3.1 Optimal Coupling and Correlation Bounds

We formalize the problem of identifying the joint distribution (coupling) of two ordinal variables with fixed marginals that maximizes or minimizes the Pearson correlation.

Proposition 3.1. *Let X and Y be discrete ordinal random variables, each with finite ordered support:*

$$X : x_1 < x_2 < \cdots < x_{K_X}, \quad Y : y_1 < y_2 < \cdots < y_{K_Y}$$

with marginal probability distributions $p_X(x_i) = P(X = x_i)$, $p_Y(y_j) = P(Y = y_j)$, respectively.

Then, the maximum and minimum values of the Pearson correlation coefficient r_{XY}, consistent with the fixed marginals, are obtained as follows:

- Maximum r_{XY}: achieved by pairing largest X-values with largest Y-values (comonotonic arrangement).

- Minimum r_{XY}: achieved by pairing largest X-values with smallest Y-values (anti-comonotonic arrangement).

Proof. The Pearson correlation coefficient is given by:

$$r_{XY} = \frac{E[XY] - E[X]E[Y]}{\sigma_X \sigma_Y}.$$

Given fixed marginals $p_X(x_i)$, $p_Y(y_j)$, the expectations $E[X], E[Y]$ and standard deviations σ_X, σ_Y are constant. Thus, to maximize or minimize r_{XY}, we only need to maximize or minimize the expectation $E[XY]$:

$$E[XY] = \sum_{i=1}^{K_X} \sum_{j=1}^{K_Y} x_i y_j P(X = x_i, Y = y_j).$$

This proof relies on a rearrangement inequality of (Hardy et al., 1952) (Theorem 368):

For real sequences $a_1 \leq a_2 \leq \cdots \leq a_n$ and $b_1 \leq b_2 \leq \cdots \leq b_n$, and for any permutation π of indices $\{1, \ldots, n\}$ we have the following:

$$a_1 b_1 + a_2 b_2 + \cdots + a_n b_n \geq a_1 b_{\pi(1)} + a_2 b_{\pi(2)} + \cdots + a_n b_{\pi(n)}$$

and

$$a_1 b_n + a_2 b_{n-1} + \cdots + a_n b_1 \leq a_1 b_{\pi(1)} + a_2 b_{\pi(2)} + \cdots + a_n b_{\pi(n)}$$

Equality occurs only if the permutation π results in a monotonically increasing sequence (for maximum) or monotonically decreasing sequence (for minimum).

To explicitly connect this to our problem, enumerate observations explicitly from sorted marginals:

$$X_{\downarrow} : x_{[K_X]} \geq x_{[K_X-1]} \geq \cdots \geq x_{[1]}, \quad Y_{\downarrow} : y_{[K_Y]} \geq y_{[K_Y-1]} \geq \cdots \geq y_{[1]}.$$

(Sorted in descending order for maximization). Next expand discrete probability distributions into explicit observation-level sequences. If $K_X \neq K_Y$, extend the shorter sequence with zero-probability categories so both sequences have equal length, which does not affect marginal probabilities or correlation.

Hence, to maximize $E[XY]$: pair the largest ranks of X with the largest ranks of Y. Explicitly, start from the top (highest values) and move downward:

- Pair as many observations of the largest X-category as possible with the largest Y-category, then next-largest, etc., until all observations are paired.
- By the Hardy–Littlewood–Pólya Rearrangement Inequality (Hardy et al., 1952), this explicitly constructed pairing yields the maximum possible sum of products, hence maximizing $E[XY]$.

Likewise, to minimize $E[XY]$ pair the largest X-categories with the smallest Y-categories, then second-largest X-categories with second-smallest Y-categories, and so forth (anti-comonotonic ordering). By the rearrangement inequality, this arrangement explicitly yields the minimal sum of products, hence minimizing $E[XY]$. ■

In words, given fixed marginals, we have shown:

- **Maximum correlation** $r_{\max}$ is achieved by comonotonic (e.g. sorting values of X and Y from largest to smallest independently and pairing) arrangement:

$$P_{\max}(X = x_{[i]}, Y = y_{[i]}) \quad \text{in either descending or ascending order.}$$

- **Minimum correlation** $r_{\min}$ is achieved by anti-comonotonic arrangement:

$$P_{\min}(X = x_{[i]}, Y = y_{[n+1-i]}) \quad \text{pairing descending X with ascending Y, or viceversa.}$$

With this explicit construction we can simply calculate r_{max} and r_{min} when the values of X and Y are ranks.

Corollary 1. (Analytic Forms for $r_{\min}$ and $r_{\max}$)

Let two ordinal variables X and Y have values as follows:

- X with K_X ordinal levels: $1, 2, \dots, K_X$

- Y with K_Y ordinal levels: $1, 2, \dots, K_Y$

with absolute frequencies:

- $n_X(i)$, number of observations at level i of X.

- $n_Y(j)$, number of observations at level j of Y.

Then the maximum Pearson correlation is:

$$r_{\max} = \frac{E[XY]_{\max} - \bar{X}\bar{Y}}{\sigma_X \sigma_Y}.$$

Where

$$E[XY]_{\max} = \frac{1}{N} \sum_{i=1}^{K_X} \sum_{j=1}^{K_Y} x_i y_j M_{i,j}$$

where explicitly defined frequencies $M_{i,j}$ pair observations in a comonotonic manner, calculated explicitly via the greedy algorithm described in the proof below.

Then explicitly the minimal expectation is:

$$E[XY]_{\min} = \frac{1}{N} \sum_{i=1}^{K_X} \sum_{j=1}^{K_Y} x_i y_j R_{i,j}$$

Where R_{ij} is defined in the proof.

Proof. Algorithm for explicit calculation of $M_{i,j}$ (greedy method):

Initialize available frequencies explicitly: $n'_Y(j) = n_Y(j) \quad \text{for each } j$

- For each level x_i, from smallest $i = 1$ to largest:
  - For each level y_j, from smallest $j = 1$ to largest:
    - Pair as many observations as possible:

$$M_{i,j} = \min \left(n_X(i) - \sum_{s=1}^{j-1} M_{i,s},\, n'_Y(j) \right)$$

Next, update explicitly:

$$n'_Y(j) \leftarrow n'_Y(j) - M_{i,j}$$

Define the matrix $M_{i,j}$, the frequency with which the level x_i pairs with the level y_j. Clearly, we must have:

$$\sum_j M_{i,j} = n_X(i), \quad \sum_i M_{i,j} = n_Y(j)$$

Similarly, the minimum Pearson correlation ($r_{\min}$) achieved by anti-comonotonic alignment (pairing largest ranks of X with smallest ranks of Y) is explicitly:

$$r_{\min} = \frac{E[XY]_{\min} - \bar{X}\bar{Y}}{\sigma_X \sigma_Y}.$$

To achieve the maximum expected product $E[XY]_{\max}$, do the following: First sort ranks from largest to smallest independently for both X and Y; then pair ranks directly from highest to lowest available observations.

Minimum Expectation

The formula for $E[XY]_{\min}$ is derived in a similar manner. Define explicitly a matrix $R_{i,j}$ giving frequencies of pairings $X = x_i$ with $Y = y_j$. For the minimal expectation, pair largest available X-values explicitly with smallest available Y-values first (greedy algorithm):

Initialize explicitly available frequencies:

$$n'_Y(j) = n_Y(j) \quad \text{for each } j$$

- For i from largest to smallest ($i = K_X$ down to 1):
  - For j from smallest to largest ($j = 1$ up to K_Y):
    - Pair explicitly as many observations as possible:

$$R_{i,j} = \min\left(n_X(i) - \sum_{s=1}^{j-1} R_{i,s},\, n'_Y(j)\right)$$

- Update explicitly available frequencies:

$$n'_Y(j) \leftarrow n'_Y(j) - R_{i,j}$$

Then explicitly the minimal expectation is:

$$E[XY]_{\min} = \frac{1}{N} \sum_{i=1}^{K_X} \sum_{j=1}^{K_Y} x_i y_j R_{i,j}$$

This explicit equation applies Whitt's rearrangement theorem , Whitt (1976) , ensuring proper maximal pairing of ranks. (Whitt (1976)) ■

Discussion

We want to construct a joint distribution table M such that:

- The **row sums** match the marginal counts for X: $\sum_j M_{i,j} = n_X(i)$
- The **column sums** match the marginal counts for Y: $\sum_i M_{i,j} = n_Y(j)$

To do this, we go **greedily**, pairing the highest available X values with the **smallest available** Y values, one cell at a time.

To avoid **exceeding** the available marginal totals in Y, we must track how much of $n_Y(j)$ is **still unassigned** at each step — and that's what $n'_Y(j)$ represents.
After assigning $M_{i,j}$ units to the cell (i, j), we subtract that amount from the remaining pool of level y_j values, like this: $n'_Y(j) \leftarrow n'_Y(j) - M_{i,j}$This ensures we don't **overfill** any column of the matrix.

To maximize the expectation $E[XY]$, the rearrangement inequality states we must pair values of X and Y from smallest-to-smallest, second-smallest-to-second-smallest, and so forth (comonotonic alignment).

Thus, explicitly:

- Sort X: lowest-to-highest:

$$X_{\text{sorted}} = (\underbrace{x_1, \dots, x_1}_{n_X(1)}, \underbrace{x_2, \dots, x_2}_{n_X(2)}, \dots, \underbrace{x_{K_X}, \dots, x_{K_X}}_{n_X(K_X)})$$

Sorted Y-values (ascending order for max, descending for min):

$$Y_{\text{sorted(max)}} = (\underbrace{y_1, \dots, y_1}_{n_Y(1)}, \underbrace{y_2, \dots, y_2}_{n_Y(2)}, \dots, \underbrace{y_{K_Y}, \dots, y_{K_Y}}_{n_Y(K_Y)})$$

$$Y_{\text{sorted(min)}} = (\underbrace{y_{K_Y}, \dots, y_{K_Y}}_{n_Y(K_Y)}, \underbrace{y_{K_Y-1}, \dots, y_{K_Y-1}}_{n_Y(K_Y-1)}, \dots, \underbrace{y_1, \dots, y_1}_{n_Y(1)})$$

The maximum and minimum expectations are simply:

Maximum expectation (pair element-by-element):

$$E[XY]_{\max} = \frac{1}{N} \sum_{k=1}^{N} X_{\text{sorted}}(k) \cdot Y_{\text{sorted(max)}}(k)$$

- **Minimum expectation**:

$$E[XY]_{\min} = \frac{1}{N} \sum_{k=1}^{N} X_{\text{sorted}}(k) \cdot Y_{\text{sorted(min)}}(k)$$

These classical bounds define feasible joint distributions given marginals and constrain the attainable values of r. There is an interesting connection to Fréchet–Hoeffding and Boole–Fréchet Inequalities, which we note but leave for future research.[1]

3.1.1 Illustration

To illustrate these idea concretely, Table 1 presents an illustrative example of comonotonic and anti-comonotonic arrangements and of the resulting lower- and upper-bounds of r, for two 4-category variables and 12 cases. One can easily expand this example to pairs of variables with different numbers of categories and different numbers of cases.

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[1]Fréchet–Hoeffding and Boole–Fréchet Inequalities: The upper bound is always attained by r_{max}. However, the lower bound is *not* guaerenteed to attain. The fundamental asymmetry arises because discrete ordinal distributions cannot smoothly bridge these gaps. Thus, the upper bound alignment is always feasible, while the lower bound alignment frequently encounters structural obstacles.

In the discrete ordinal setting, r_{max} is **always** attained by the FH upper (comonotonic) construction.
(BFbounds.md; Proofs-OrdinalVariables.md; Ordinal Variables and Correlation.md)

The FH lower (countermonotonic) construction may be **unattainable** in discrete settings; r_{min} is then the best feasible opposite-order arrangement.
(BFbounds.md; Proofs-OrdinalVariables.md; Ordinal Variables and Correlation.md)

Upper bound (positive alignment):

- Sorting X and Y separately in descending order, then pairing largest-to-largest, smallest-to-smallest, **always works**.
- You never run out of "matching" observations because the descending order for both variables naturally aligns category sizes. Even if one variable has large gaps, you can still sequentially pair without difficulty.

Lower bound (negative alignment):

- You sort X descending but Y ascending. You're pairing extremes together: highest X with lowest Y, lowest X with highest Y.
- Now you face potential "imbalance":
  - If one category is large for X but the corresponding opposite-ranked category for Y is small, you quickly exhaust available "opposite-end" observations.
  - After exhausting the best opposites, you are forced to pair less "extreme" pairs, creating a suboptimal alignment.
- Hence, perfect negative alignment (lower bound) becomes impossible to achieve fully, leaving you short of the theoretical Fréchet–Hoeffding lower bound.

Table 1: Determining minimum and maximum attainable values of correlation between two ordered categorical variables X and Y given their marginal distributions.

(a) Observed frequency distributions

	Category 1	**Category 2**	**Category 3**	**Category 4**	**Total**
X	6	3	2	1	12
Y	3	5	2	2	12

(b) Comonotonic arrangement and pairing:

X: 1 1 1 1 1 1 2 2 2 3 3 4

Y: 1 1 1 2 2 2 2 2 3 3 4 4 $r_{\text{max}} = 0.87831$

(c) Anti-comonotonic arrangement and pairing:

X: 1 1 1 1 1 1 2 2 2 3 3 4

Y: 4 4 3 3 2 2 2 2 2 1 1 1 $r_{\text{min}} = -0.79466$

Table 2: Illustrative pairs of marginal distributions and their associated minimum and maximum attainable values of r (r_{min}, r_{max}) given these distributions. The distributions are given in counts (n=12). The categories are assigned consecutive integer values starting with 1. C1 and C2 are measures of constraint, as discussed in the text.

	Category				**Total**	r_{min}	r_{max}
	1	**2**	**3**	**4**			
(a) Two arbitrarily distributed ordinal categorical variables (orcats)							
X	6	3	2	1	12	-0.79466	0.87831
Y	3	5	2	2	12		
(b) One arbitrarily distributed variable, and one with same distribution inversed							
X	6	3	2	1	12	-1	0.71429
Y	1	2	3	6	12		
(c) Two variables with arbitrary identical distributions							
X	6	3	2	1	12	-0.71429	1
Y	6	3	2	1	12		
(d) Two variables with identical symmetric and unimodal distributions							
X	2	4	4	2	12	-1	1
Y	2	4	4	2	12		
(e) Two variables with different distributions, both symmetric and unimodal							
X	2	4	4	2	12	-0.91169	0.91169
Y	1	5	5	1	12		
(f) Two variables with identical symmetric bimodal distributions							
X	4	2	2	4	12	-1	1
Y	4	2	2	4	12		
(g) Two variables with different distributions, both symmetric and bimodal							
X	4	2	2	4	12	-0.95673	0.95673
Y	5	1	1	5	12		
(h) One symmetric unimodal variable, the other symmetric bimodal							
X	2	4	4	2	12	-0.89923	0.89923
Y	4	2	2	4	12		
(i) Two variables, one very left skewed, the other very right skewed							
X	1	1	1	9	12	-0.95818	0.31939
Y	8	2	1	1	12		
(j) Same as situation (i), above, but categories assigned values 1; 2.25; 2.75; 4							
X	1	1	1	9	12	-0.93321	0.33829
Y	8	2	1	1	12		

3.2 Examples

Below, "scores 0:$(K-1)$" means category values $[0, 1, \ldots, K-1]$. "Affine recode" means $a + b \times$ (old score) with $b > 0$. "Non-affine monotone" means strictly increasing but not affine (e.g., convex spacing).

3.2.1 Notation (permutation extremes, "Lane A")

- Build expanded vectors x and y from **counts** and **scores**.
- $r_{\max} = \operatorname{cor}(\operatorname{sort}(x, \downarrow), \operatorname{sort}(y, \downarrow))$.
- $r_{\min} = \operatorname{cor}(\operatorname{sort}(x, \downarrow), \operatorname{sort}(y, \uparrow))$.

3.2.2 A) Same number of categories, but $|r_{\min}| \neq r_{\max}$ (concrete numbers)

Counts (both

$$K = 5$$

):

- $$X: \ [100,\ 200,\ 300,\ 200,\ 200]$$
- $Y: \ [100,\ 100,\ 100,\ 300,\ 400]$ (skewed; not palindromic)

Scores: 0:4 for both.

- $$r_{\max} \approx 0.929398758$$
- $$r_{\min} \approx -0.881118303$$
- $$|r_{\min}| \approx 0.8811 \neq r_{\max}$$

Affine recode (10,20,30,40,50) for both:

- $r_{\max}, r_{\min}$ **unchanged** (up to rounding).

Non-affine monotone recode for Y: $[0, 1, 1.5, 3, 10]$.

- $$r_{\max} \approx 0.930261031$$
- $$r_{\min} \approx -0.863250703$$
- Values change (Pearson is **not** invariant to non-affine spacing); magnitudes still unequal.

Why unequal magnitudes? No palindromic symmetry in Y the "palindrome-sum" pattern in y'–space isn't zero, so $S_{\max} + S_{\min} \neq 0$.

3.2.3 B) Different numbers of categories, yet $|r_{\min}| = r_{\max}$ (concrete numbers)

Counts:

- X (4 categories): [100, 200, 300, 400]
- Y (5 categories): [100, 200, 400, 200, 100] (**palindromic**)

Scores: $X:$ 0:3, $Y:$ 0:4.

- $$r_{\max} \approx 0.912870929$$
- $$r_{\min} \approx -0.912870929$$
- $|r_{\min}| = r_{\max}$ (to machine precision)

Affine recode (e.g., 10,20,...):

- Magnitudes remain **identical**; affine transforms don't change Pearson correlation.

Non-affine monotone recode for Y: $[0, 1, 1.5, 3, 10]$.

- $$r_{\max} \approx 0.580072921$$
- $$r_{\min} \approx -0.842041337$$
- **Magnitudes no longer match** because antisymmetry of the **scores** about the center is lost, even though the **counts** are palindromic.

Why equal magnitudes in the affine case? Palindromic **mass** symmetry in Y + **antisymmetric scoring** (any affine map of 0:$(K_Y - 1)$) the palindrome-sum vector $y'_i + y'_{n+1-i} \equiv 0$. Hence $S_{\max} + S_{\min} = 0$ $|r_{\min}| = r_{\max}$ for **any** X, irrespective of $K_X \neq K_Y$.

3.2.4 C) Spearman invariance (same data as in B)

Compute ranks (with average ties) **once** for x and y, then form extremes by sorting those rank vectors.

- With Y scored 0:4:
  $r_{\max}^{\text{Spearman}} \approx 0.920837215, \quad r_{\min}^{\text{Spearman}} \approx -0.920837215.$
- With Y scored non-affinely $[0, 1, 1.5, 3, 10]$:
  identical Spearman extremes: 0.920837215 and −0.920837215.

Spearman is invariant to any strictly monotone recode because ranks depend only on order, not spacing.

3.2.5 What Changes and What Doesn't (Pearson vs. Spearman)

- **Pearson:**
  - **Affine** recodes $a + b \cdot \text{score}$ (with $b > 0$) leave $r_{\max}, r_{\min}$, and any $|r_{\min}| = r_{\max}$ conclusions **unchanged**.
  - **Non-affine monotone** recodes can change $r_{\max}, r_{\min}$ and can break $|r_{\min}| = r_{\max}$ even with palindromic counts.
- **Spearman:**
  - Any strictly monotone recode leaves extremes **unchanged**.
  - Palindromic counts in Y (with the usual symmetric rank scheme) give $|r_{\min}| = r_{\max}$.

3.3 Population Bounds

In this section we further explore the theoretical properties on r_{min} and r_{max}.

3.3.1 Lemma (signs of extreme correlations under monotone scoring)

Lemma 1. Let X and Y be ordinal variables. Assign numeric scores to categories via **non-decreasing** maps s_X, s_Y that respect the category order (scores may be affine or nonlinear; ties allowed). Define two extreme couplings:

- **Comonotone (upper)**: pair larger scored values of X with larger scored values of Y. - **Antitone (lower)**: pair larger scored values of X with smaller scored values of Y.

Let $r_{\max}$ and $r_{\min}$ be the Pearson correlations computed from these couplings, either (i) in the **finite-sample/permutation** sense (sort scored observations in the same vs. opposite order), or (ii) in the **Fréchet–Hoeffding (FH)/population** sense (pair quantiles $Q_X(u)$ with $Q_Y(u)$ vs. $Q_Y(1-u)$).

Then:

1. $r_{\max} \geq 0$ and $r_{\min} \leq 0$.

2. Moreover, $r_{\max} > 0$ and $r_{\min} < 0$ **iff** both variables have **positive variance** under the chosen scoring (i.e., neither scored variable is almost surely constant).

(If you reverse the scoring of one variable (decreasing map), swap "upper" and "lower". For Spearman's ρ, take the scores to be midranks—monotone by construction—and the same conclusions hold.)

Proof. Proof

We give parallel proofs in the finite-sample and population settings.

A) Finite-sample (permutation) setting

Let $x_{(1)} \geq \cdots \geq x_{(n)}$ and $y_{(1)} \geq \cdots \geq y_{(n)}$ be the **scored** observations of X and Y sorted in nonincreasing order (monotone scoring ensures sorting by category equals sorting by score). Center them:

$$x'_i := x_{(i)} - \bar{x}, \qquad y'_i := y_{(i)} - \bar{y}, \quad \bar{x} := \tfrac{1}{n}\sum_i x_{(i)},\ \bar{y} := \tfrac{1}{n}\sum_i y_{(i)}.$$

Define the two extreme centered sums

$$S_{\max} := \sum_{i=1}^{n} x'_i\, y'_i, \qquad S_{\min} := \sum_{i=1}^{n} x'_i\, y'_{n+1-i}.$$

- **Signs.**
  Chebyshev's sum inequality (rearrangement for similarly/ oppositely ordered sequences) gives

  $$\frac{1}{n}\sum_i x'_i y'_i \ \geq\ \Big(\tfrac{1}{n}\sum_i x'_i\Big)\Big(\tfrac{1}{n}\sum_i y'_i\Big) = 0, \quad \Rightarrow\ S_{\max} \geq 0,$$

  and, because y'_{n+1-i} is **nondecreasing** in i,

  $$\frac{1}{n}\sum_i x'_i y'_{n+1-i} \ \leq\ \Big(\tfrac{1}{n}\sum_i x'_i\Big)\Big(\tfrac{1}{n}\sum_i y'_{n+1-i}\Big) = 0, \quad \Rightarrow\ S_{\min} \leq 0.$$

  Dividing by $\sigma_X\sigma_Y > 0$ (the standard deviations of the scored variables) yields $r_{\max} \geq 0$ and $r_{\min} \leq 0$.

- **Strictness ($\Leftrightarrow$ positive variances).**
  Suppose $\sigma_X > 0$ and $\sigma_Y > 0$ (i.e., neither $\{x'_i\}$ nor $\{y'_i\}$ is identically zero). The rearrangement inequality says $S_{\max}$ is the **maximum** and $S_{\min}$ the **minimum** of $\sum_i x'_i y'_{\pi(i)}$ over all permutations π; moreover $S_{\min} < S_{\max}$. The **average** over all permutations equals

  $$\frac{1}{n!}\sum_\pi \sum_i x'_i y'_{\pi(i)} = \sum_i x'_i\Big(\frac{1}{n!}\sum_\pi y'_{\pi(i)}\Big) = \sum_i x'_i \cdot \frac{1}{n}\sum_j y'_j = 0.$$

  Since the minimum is < the maximum and the average is 0, it follows that $S_{\min} < 0 < S_{\max}$. Hence $r_{\min} < 0 < r_{\max}$.

  Conversely, if $\sigma_X = 0$ or $\sigma_Y = 0$, then one centered sequence is identically zero, so $S_{\max} = S_{\min} = 0$ and the correlations are undefined or (in a limiting sense) attain 0. This matches the "only if" direction.

B) Population (FH/quantile) setting

Let $Q_X, Q_Y : [0,1] \to \mathbb{R}$ be the quantile functions of the **scored** variables. Set

$$f(u) := Q_X(u) - \mu_X, \qquad g(u) := Q_Y(u) - \mu_Y,$$

so $\int_0^1 f = \int_0^1 g = 0$. Monotone scoring preserves category order, hence Q_X, Q_Y are nondecreasing; therefore f, g are nondecreasing functions.

- **Upper/comonotone covariance**:

$$\mathrm{Cov}_{\max} = \int_0^1 f(u)\, g(u)\, du \;\geq\; \Big(\int_0^1 f\Big)\Big(\int_0^1 g\Big) = 0$$

  by Chebyshev's **integral** inequality for similarly ordered functions.

- **Lower/antitone covariance**:

$$\mathrm{Cov}_{\min} = \int_0^1 f(u)\, g(1-u)\, du \;\leq\; \Big(\int_0^1 f\Big)\Big(\int_0^1 g(1-u)\Big) = 0,$$

  since f is nondecreasing and $g(1-u)$ is **nonincreasing**.

Dividing by $\sigma_X \sigma_Y > 0$ yields $r_{\max} \geq 0$ and $r_{\min} \leq 0$.

- **Strictness.**
  If both variables have positive variance, then f and g are not a.e. constant. For nontrivial nondecreasing f, g, Chebyshev's integral inequalities above are **strict** unless one function is a.e. constant. Hence $\mathrm{Cov}_{\max} > 0$ and $\mathrm{Cov}_{\min} < 0$, giving $r_{\max} > 0$ and $r_{\min} < 0$. Conversely, if either variance is 0, both covariances are 0.

This completes the proof. □ ■

3.4 Asymetry

Symmetry and the Role of Ties

- Conditions under which $r_{\min} = -r_{\max}$

PROP: If one marginal is **palindromically symmetric** about its center rank, then $r_{\min} = -r_{\max}$.
(Cees – r_min MAGNITUDE Asymmetry in Pearson Correlation.md)

Necessary/sufficient symmetry condition (precisely stated)

- **Pair-specific equality** $|r_{\min}| = r_{\max} \iff \sum_i x'_i\,(y'_i + y'_{n+1-i}) = 0$.
  (Can happen by coincidence even if Y isn't symmetric.)
- **Universal in** X equality (for **all** X):
  Necessary and sufficient that **one** marginal (say Y) has

(i) **palindromic mass symmetry** $p_Y(k) = p_Y(K_Y - 1 - k)$ and

(ii) **antisymmetric scoring** $s_Y(k) + s_Y(K_Y - 1 - k) = \text{const}$ (true for any affine map of $0{:}(K_Y - 1)$).
This condition is **independent of K_X vs K_Y**.

3.4.1 1) setup (permutation extremes for a single sample)

Let $x_{(1)} \geq \cdots \geq x_{(n)}$ and $y_{(1)} \geq \cdots \geq y_{(n)}$ be the sorted sample values (or expanded from counts). Define centered sequences

$$x'_i := x_{(i)} - \bar{x}, \qquad y'_i := y_{(i)} - \bar{y},$$

and the two permutation extremes

$$S_{\max} = \sum_{i=1}^{n} x'_i \, y'_i, \qquad S_{\min} = \sum_{i=1}^{n} x'_i \, y'_{n+1-i},$$

with $r_{\max} = S_{\max}/(\sigma_X \sigma_Y)$, $r_{\min} = S_{\min}/(\sigma_X \sigma_Y)$.

We always have $S_{\max} \geq 0$ and $S_{\min} \leq 0$. Magnitude symmetry $|r_{\min}| = r_{\max}$ is equivalent to

$$S_{\max} + S_{\min} = 0.$$

3.4.2 2) pair-specific necessary & sufficient condition

Let

$$z_i := y'_i + y'_{n+1-i}.$$

Then

$$S_{\max} + S_{\min} = \sum_{i=1}^{n} x'_i \, z_i.$$

Hence, for this *particular* pair (x, y),

$$\boxed{|r_{\min}| = r_{\max} \iff \sum_i x'_i \, z_i = 0} \tag{1}$$

(i.e., the centered X sequence is orthogonal to the “palindrome-sum” vector z).
This can hold even if Y is not symmetric—by coincidence of that inner product being zero.

3.4.3 3) universal condition (independent of X)

If you want $|r_{\min}| = r_{\max}$ to hold for **every** possible X (every centered, sorted x'), then Equation 1 must hold for all x'. The only way is

$$z_i \equiv 0 \quad \text{for all } i,$$

i.e.

$$y'_i + y'_{n+1-i} \equiv 0 \quad \forall i.$$

Translating back to categories and scores, a **necessary and sufficient universal condition** is:

- **Palindromic mass symmetry for Y:** $p_Y(k) = p_Y(K_Y - 1 - k)$ for all k, and
- **Antisymmetric scoring for Y:** the score map satisfies $s_Y(k) + s_Y(K_Y - 1 - k)$ is constant (true for affine scores like $0, 1, \dots, K_Y - 1$, and for Spearman midranks under palindromic counts).

Under these, $z_i \equiv 0$ and thus $|r_{\min}| = r_{\max}$ for any X.

> If either mass symmetry or score antisymmetry fails, you can generally find an X for which $|r_{\min}| \neq r_{\max}$ (often $|r_{\min}| > r_{\max}$ when Y is skewed).

- How ties in marginal distributions affect the attainability of the lower bound

1. $r_{\max}$ **= FH upper (comonotonic), always attainable** in discrete ordinal settings.
2. $r_{\min}$ may **not** equal FH lower in discrete settings; use best feasible opposite-order arrangement.
3. No universal bound on relative magnitudes: $|r_{\min}|$ may be less than, equal to, or greater than $r_{\max}$.
4. Palindromic symmetry of one marginal guarantees $|r_{\min}| = r_{\max}|$.

The universal condition established above guarantees $|r_{\min}| = r_{\max}$ when one marginal is palindromically symmetric with antisymmetric scoring. When that condition fails, the bounds can be asymmetric in either direction. The following corollary addresses the practically important case in which $|r_{\min}|$ strictly exceeds $r_{\max}$.

Corollary 2. (Strict Asymmetry of Bounds)

Let X and Y be discrete ordinal variables with positive variance under consecutive-integer scoring. Then $|r_{\min}| > r_{\max}$ is possible, and occurs if and only if

$$S_{\max} + S_{\min} < 0,$$

where $S_{\max} = \sum_i x'_i y'_i$ and $S_{\min} = \sum_i x'_i y'_{n+1-i}$, equivalently $\sum_i x'_i z_i < 0$ with $z_i := y'_i + y'_{n+1-i}$. This condition holds when a marginal is sufficiently skewed that anti-comonotonic pairing generates a larger absolute sum of products than comonotonic pairing.

Proof. Since $\sigma_X \sigma_Y > 0$, we have $|r_{\min}| > r_{\max} \iff |S_{\min}| > S_{\max} \iff S_{\max} + S_{\min} < 0$, using $S_{\min} \leq 0$ and $S_{\max} \geq 0$. Since $S_{\max} + S_{\min} = \sum_i x_i' z_i$, the algebraic condition follows directly from the pair-specific condition ($\star$).

The mechanism is as follows. When Y is right-skewed, most probability mass sits in low categories but the distribution has a long upper tail. Its centered values y_i' are therefore large in magnitude at high categories and near zero at low categories. Under anti-comonotonic pairing, the large positive values of X are matched against the small centered values of Y at low categories, while the large negative values of X are matched against the large positive tail of Y. This asymmetric allocation of centered values means $|S_{\min}|$ can exceed $S_{\max}$, producing $|r_{\min}| > r_{\max}$. $\square$ ■

Example 1 (BES 2019: strict asymmetry of bounds)**.** The British Election Study 2019 contains a variable pair in which X has $K_X = 4$ ordered categories and Y has $K_Y = 5$ ordered categories, with both marginals skewed. For this pair, the comonotonic construction yields $r_{\max} = 0.850$ while the anti-comonotonic construction yields $r_{\min} = -0.899$, so $|r_{\min}| > r_{\max}$ by a margin of 0.049. The attainable interval $[-0.899, 0.850]$ is not centred at zero. This example confirms that strict bound asymmetry occurs in real survey data and that assuming symmetry would misrepresent the feasible range of r.

3.5 Estiimation and Joint Uncertainty

VH

4 Measurement Issues and Metrics of Constraint

IN THIS SECTION: "How to define metrics (on *pairs* of discrete probability distributions) to *explain*:

VH

1.
$$\frac{r_{max} - r_{min}}{2}$$

2.
$$1 - r_{max}$$

3.
$$|-1 - r_{min}|$$

Or analogs for $\hat{r}^2$:

1.
$$\frac{r_{max}^2 - r_{min}^2}{2}$$

1a.
$$\frac{(r_{max} - r_{min})^2}{2}$$

2.

$$1 - r_{max}^2$$

3.

$$1 - r_{min}^2$$

As demonstrated above, the extent to which marginal distributions of orcats constrain the range of attainable value of r varies considerably. In the few illustrative examples presented in Table 2 rmin varied between -1 and -0.71, while rmax varied between 0.32 and 1. Moreover, in Table 2 rmin and rmax do not vary in lockstep but at least to some extent independent of each other.[fn: Table 2 is a poor basis to assess the functional relationship between rmin and rmax as it contains a small number of handpicked situations. This relationship will be investigated in detail later, in section ??.] This raises the question how strong the overall constraining effect is that marginal distributions exert on the attainable values of r. It also raises the question of what factors affect the strength of this overall constraining effect. In this section we describe two ways in which the strength of constraints can be defined. Later (in section ??) we turn to the questions of the conditions that drive the strength of this effect. Perhaps the most obvious way to express the strength of the marginal constraints on the attainable values of r is by comparing the width of the interval between rmin and rmax with that of the ubiquitously used reference interval [-1, 1]. As the width of the latter interval is 2, it is advantageous to express the strength of the overall constraining effect –to be denoted as C_1 as

$$C_1 = \frac{|r_{min}| + |r_{max}|}{2}$$

C_1 reflects the proportion of the range of the attainable values of r compared to the theoretically maximum width of this range. Applying this measure to the illustrative cases listed in Table 2 shows that C1 varies. Its highest value is 1, reflecting the absence of any constraining effect (see examples (d) and (f) in Table 2). The lowest value of C1 in Table 2 is for example (i), where it is 0.639. In other words, the marginal distributions of the two variables in that example restrict the range of its attainable values to 63.9% of the traditional [-1, 1] reference interval. Some might object to the C_1 measure on the grounds that it assumes that r (and thus also rmin and rmax) is expressed at interval level and that one could thus state validly that a correlation of 0.4 reflects an association twice as strong as one of 0.2. That would be so if numerical differences between correlations correspond to similar differences in observable aspects of data [insert references to correspondence NRS to ERS]. That not being the case, one could instead compare r2s –percentages of shared variance– rather than rs. [fn: This perspective is not uncontested, however, and its usefulness depends on the purpose of inspecting correlations: as reflections of the association between variables, or as reflections of associations between indicators of a latent concept (for an elaboration of this difference, cf. Ozer 1985).] This would lead to a somewhat different measure of the strength of the constraining effect of marginal distributions on the attainable values of r, as follows:

$$C_2 = \frac{r_{min}^2 + r_{max}^2}{2}$$

For the examples in Table 2, the values C2 are, respectively, 0.701; 0.755; 0.715; 1; 0.831; 1; 0.809; 0.915 and 0.510. Obviously, C1 and C2 are correlated, but how strongly cannot be assessed very well from the few handpicked examples in Table 2, but it will be considered below, using more encompassing sets of data.

4.1 Possible "Metrics of Constraint"

4.1.1 Overlap-Based Measures of Asymmetry

In addition to Total Variation (TV) distance, other metrics can quantify the structural dissimilarity between a marginal distribution and its reverse. One natural approach is to assess the degree of **overlap** between the original distribution p and its reverse p^{rev}, defined by reversing the order of the categories.

We consider three complementary measures of overlap and their complements:

4.1.1.1 (i) Total Variation (TV) Distance

The TV distance, already introduced in Section 5.5, quantifies the maximum absolute discrepancy between the two distributions:

$$\text{TV}(p, p^{\text{rev}}) = \frac{1}{2} \sum_{i=1}^{K} |p_i - p_{K+1-i}| \,.$$

4.1.1.2 (ii) Bhattacharyya Coefficient (BC) and its Complement

The **Bhattacharyya Coefficient** provides a geometric measure of similarity:

$$\text{BC}(p, p^{\text{rev}}) = \sum_{i=1}^{K} \sqrt{p_i \cdot p_{K+1-i}}.$$

Its complement, $1 - \text{BC}$, can be interpreted as a divergence-like measure that reflects geometric differences between the original and reversed distributions. While not a metric, it is bounded between 0 and 1, with 0 indicating perfect symmetry.

4.1.1.3 (iii) Overlap Coefficient (OVL)

The **Overlap Coefficient**, sometimes referred to as the Szymkiewicz–Simpson coefficient, captures the shared probability mass:

$$\text{OVL}(p, p^{\text{rev}}) = \sum_{i=1}^{K} \min\{p_i, p_{K+1-i}\}.$$

The complement $1 - \text{OVL}$ thus reflects the fraction of mass that is not shared between the two distributions.

These measures differ in their interpretive emphasis: - TV focuses on the total discrepancy. - BC captures geometric congruence. - OVL captures the total shared area between the distributions.

In our simulation study (Section 5), we compute all three diagnostics to illustrate how marginal asymmetry correlates with asymmetry in achievable Pearson correlation bounds.

Empirically, we find that all three metrics provide consistent, interpretable diagnostics of distributional asymmetry. Using them jointly provides a robust triangulation of symmetry breaking, especially in applied contexts where ordinal variables are discretized from latent traits or ratings.

4.1.1.4 (iv) Shannon Entropy

The three overlap-based measures above each compare p against its reversal p^{rev}, diagnosing whether a distribution is *internally palindromic*. A complementary perspective asks not whether a distribution is symmetric with respect to its own reversal, but how *concentrated* it is overall. Shannon entropy answers this question.

For a marginal distribution $p = (p_1, \dots, p_K)$, the Shannon entropy is:

$$H(p) = -\sum_{i=1}^{K} p_i \log p_i,$$

with the convention $0 \log 0 = 0$. Entropy is maximised at $H = \log K$ when p is uniform and minimised at $H = 0$ when all mass falls on a single category. As entropy decreases, the distribution becomes more concentrated, and — as the asymmetry results in Section 3 show — concentrated (skewed) marginals are associated both with more severe constraint on r and with greater bound asymmetry $|r_{\min}| - r_{\max}|$.

For a pair (X, Y), we summarise marginal concentration via the mean entropy:

$$\bar{H}(X, Y) = \frac{H(p_X) + H(p_Y)}{2}.$$

Low $\bar{H}$ (concentrated marginals) predicts small C_1 (narrow attainable range) and larger $\big||r_{\min}| - r_{\max}\big|$ (greater bound asymmetry). This relationship is examined empirically in Section 5 and Section 6.

The entropy measure complements rather than replaces TV, BC, and OVL: where the overlap measures diagnose *whether* a given distribution is internally symmetric, entropy diagnoses *how concentrated* it is in absolute terms. The two perspectives are related but distinct — a highly concentrated distribution tends to be non-palindromic, but palindromic distributions can also be highly concentrated (e.g., all mass at the midpoint category).

5 Monte Carlo Validation

To investigate factors that affect the bounds of attainable values of r and thus the strength of constraints exerted by marginal distributions, we construct a large simulation that spans the space of possible marginal shapes for orcats with 4, 5, 6, 7, 10, and 11 categories — the most common category counts in survey research. For each combination of category counts (K_1, K_2) we draw 2,000 pairs of marginal distributions independently and at random from the uniform Dirichlet distribution, Dirichlet($\mathbf{1}$). This prior is maximally uninformative over the simplex: it assigns equal probability to all marginal shapes, from nearly uniform to highly concentrated, without imposing any assumptions about unimodality, skewness, or location. The simulation therefore constitutes a near-exhaustive census of possible orcat pair configurations for the category counts studied. For each of the $6 \times 6 = 36$ category-count combinations we generate 2,000 pairs, yielding a simulation dataset of 72,000 orcat pairs in total. Pairs in which either variable has zero variance are excluded (none arose in practice). For each pair we compute $r_{\min}$, $r_{\max}$, C_1, C_2, and the constraint metrics defined in Section 4. Further details of the simulation procedure are provided in Appendix C.

This simulated dataset constitutes, within the limits of granularity imposed by the discrete category structure, a random sample from the full space of possible marginal distribution pairs for each (K_1, K_2) combination. It is therefore well suited to investigating the functional relationships between marginal shape and the attainable range of r. It is not, however, a random sample from the pairs of orcats that applied researchers encounter in practice: survey designers systematically avoid items that produce near-zero variance in respondents' answers. The British Election Study 2019 data analysed in Section 6 provides the empirical complement, characterising constraint severity in a real, large-scale survey instrument.

5.1 Effect of Number of Categories on Constraint Severity

The most direct driver of how tightly the attainable range of r is constrained is the number of categories K. Figure 1 shows the distribution of C_1 separately for each of the six $K \times K$ configurations, where K is the same for both variables in a pair.

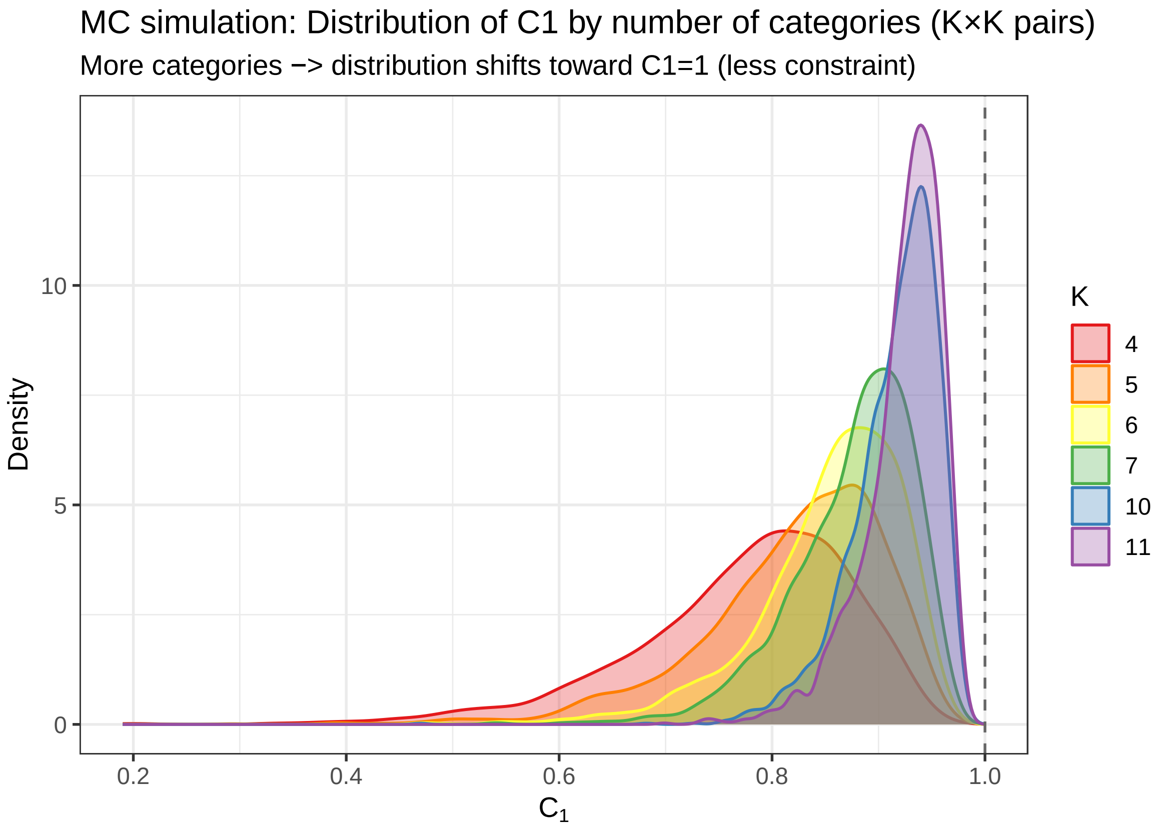

Figure 1: Distribution of constraint severity (C_1) across 2,000 simulated orcat pairs for each number of categories K, where both variables in a pair share the same K. Vertical dashed line at $C_1 = 1$ marks the unconstrained reference. Marginal distributions are drawn independently from Dirichlet($\mathbf{1}$), which is uninformative over the simplex.

The pattern is unambiguous. For $K = 4$, the distribution of C_1 is broad and left-skewed, with substantial density below 0.6 and a mode around 0.75: four-category orcats face a wide range of constraint severity, and many configurations are severely constrained. As K increases from 4 to 11, the distributions shift progressively rightward and become more tightly concentrated near $C_1 = 1$. By $K = 11$, most pairs have $C_1 > 0.90$ and the distribution has a sharp peak near 0.95.

The mechanism is straightforward. With only four categories and 1,000 observations, a single skewed draw from the Dirichlet can place a large fraction of mass in one or two categories, severely restricting what joint distributions are possible and consequently what correlation values are attainable. With eleven categories, random draws from the Dirichlet are more dispersed on average — the law of large numbers for the simplex means that the effective concentration of any one category is smaller — and the resulting marginals allow a wider range of joint distributions.

This finding has a direct practical implication: survey batteries that mix 4-point and 11-point items are mixing pairs whose attainable correlation ranges differ systematically and substantially. Two pairs each with an observed $r = 0.40$ are not comparable if one involves 4-category items and the other 11-category items.

5.2 Entropy as a Predictor of Constraint Severity

The number of categories is a coarse summary of constraint risk. Section 4 introduced Shannon entropy $\bar{H}(X, Y) = (H(p_X)+H(p_Y))/2$ as a finer, pair-specific measure of marginal concentration. Figure 2 shows the relationship between $\bar{H}$ and C_1 across the full simulation.

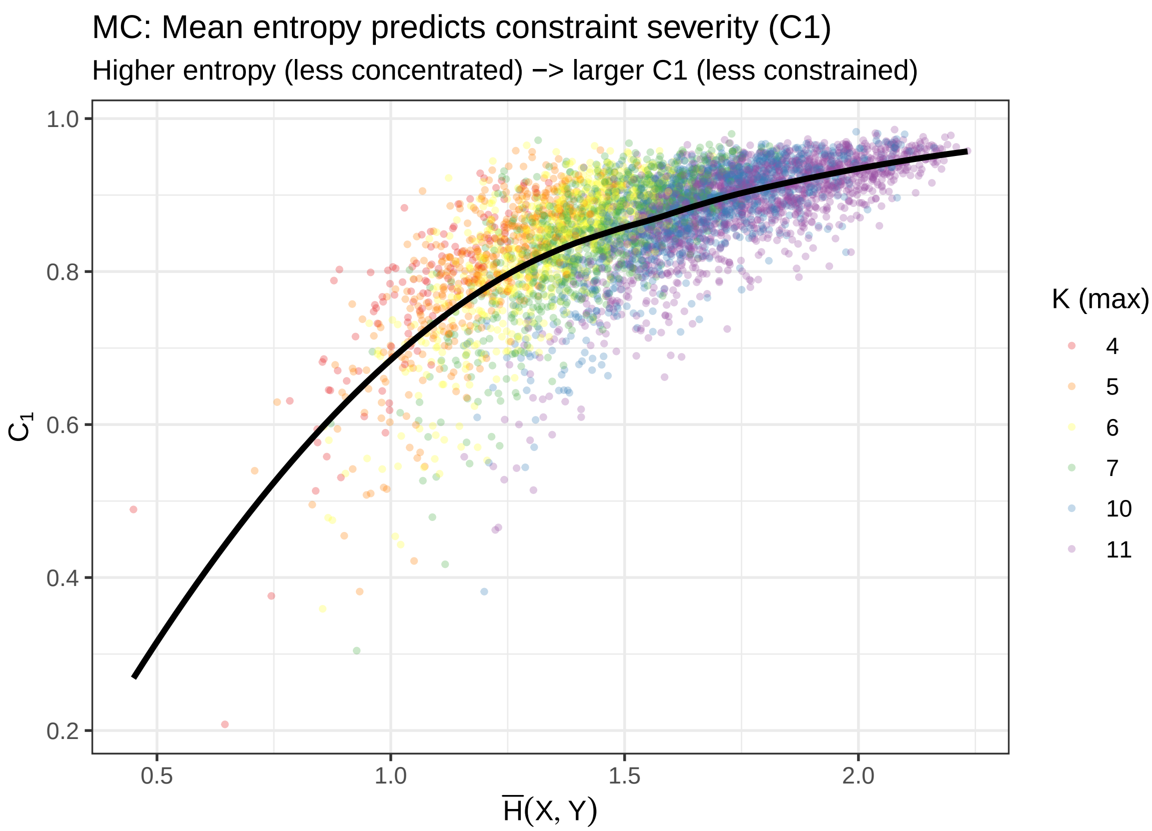

Figure 2: Scatterplot of mean Shannon entropy $\bar{H}(X, Y)$ against constraint severity C_1 for 5,000 randomly sampled simulation pairs (coloured by maximum category count K). The solid black curve is a loess smooth fitted to all 72,000 pairs.

The relationship is strongly positive and concave. At low entropy ($\bar{H} \lesssim 0.8$), both variables are highly concentrated — most probability mass sits in one or two categories — and C_1 can fall as low as 0.2. At high entropy ($\bar{H} \gtrsim 1.8$), the marginals are nearly uniform and C_1 approaches 1. The concavity reflects diminishing returns: moving from moderate to high entropy produces progressively smaller gains in the attainable range.

The colour coding by K reveals why the number of categories matters: higher K unlocks higher entropy values (the maximum entropy of a K-category uniform distribution is $\log K$, which grows with K), so the right-hand portion of the horizontal axis is populated mainly by high-K pairs. But within any K value, entropy still varies substantially depending on the specific marginal shape drawn, and it is this within-K variation that entropy captures over and above the coarser K-count predictor.

5.3 Bound Asymmetry: Direction Is Not Predictable

The theoretical results in Section 3 establish precisely when $|r_{\min}| > r_{\max}$ (the strict asymmetry corollary), and the condition involves the interaction of both marginals through the inner product $\sum_i x'_i z_i$. One might expect that variables with more concentrated or skewed marginals would systematically exhibit $|r_{\min}| > r_{\max}$. The simulation shows this expectation is only weakly supported.

Across 72,000 simulated pairs, 36,125 (50.2%) have $|r_{\min}| > r_{\max}$ and 35,875 (49.8%) have $r_{\max} > |r_{\min}|$: the split is almost exactly equal. Examining the relationship between $\bar{H}$ and asymmetry $|r_{\min}| - r_{\max}$ reveals a very slight positive tendency at low entropy (concentrated marginals are marginally more likely to produce $|r_{\min}| > r_{\max}$), but the loess smooth barely departs from zero across the entire entropy range and the scatter is enormous. Shannon entropy, total variation distance, and the other single-variable metrics in Section 4 all have negligible predictive power for the *direction* of asymmetry.

This is not a failure of the theory. The sign of $S_{\max} + S_{\min} = \sum_i x'_i z_i$ depends on the full joint structure of how both sorted marginals interact — it cannot be reduced to univariate summary statistics of each marginal separately. The practical upshot is that bound asymmetry is pair-specific and must be computed directly rather than inferred from marginal summaries: there is no shortcut. The prevalence of asymmetry — roughly half of all pairs — confirms that assuming $|r_{\min}| = r_{\max}$ is, on average, wrong for half of any given collection of orcat pairs.

While the *direction* of asymmetry is unpredictable, its *magnitude* is not. Figure 3 shows the distribution of $\big||r_{\min}| - r_{\max}\big|$ for $K \times K$ pairs across the six category counts.

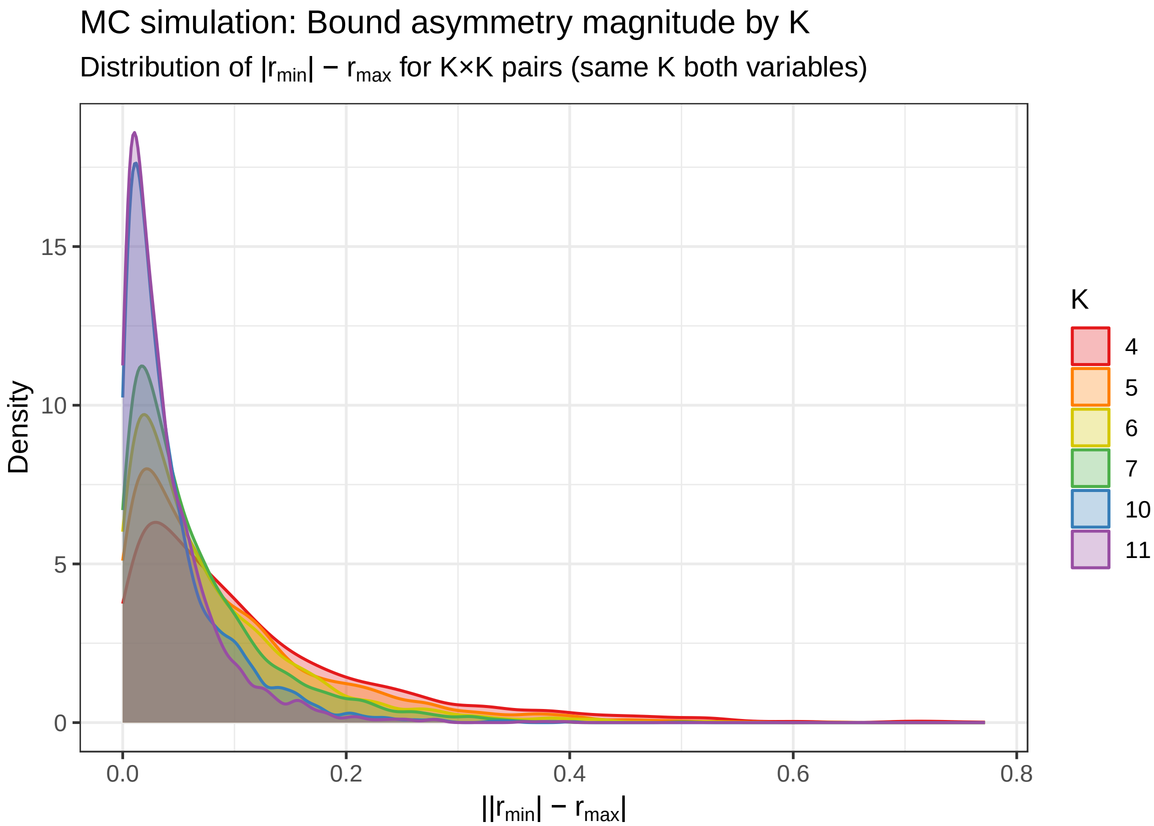

Figure 3: Distribution of bound asymmetry magnitude $||r_{\min}| - r_{\max}|$ for simulated $K \times K$ orcat pairs (both variables sharing the same K), across the six category counts. All distributions are right-skewed with a mode near zero, but the right tail shrinks systematically as K increases.

The distributions shift systematically leftward as K increases. At $K = 4$, the median asymmetry magnitude is 0.079 and the distribution has a heavy right tail extending to 0.6: a typical pair of 4-category orcats exhibits bounds differing in magnitude by around 8 correlation units, with some pairs differing by far more. At $K = 11$, the median falls to 0.027 and the right tail is negligible. The full progression — medians of 0.079, 0.062, 0.051, 0.044, 0.028, and 0.027 for $K = 4, 5, 6, 7, 10, 11$ respectively — is strictly monotone decreasing.

The practical implication complements the direction finding. An analyst who assumes $|r_{\min}| = r_{\max}$ for a pair of 4-category orcats is, on average, wrong by about 0.08 in magnitude — not trivial for an effect-size comparison or a hypothesis test calibrated to bounds. For 11-category items the error is smaller, around 0.03 on average. Neither is zero, but the severity of the approximation error depends strongly on the scale resolution of the items involved.

6 Empirical Illustration: BES 2019

The British Election Study 2019 (BES 2019) is a large-scale post-election survey of British voters, designed to measure political attitudes, values, party identification, electoral behaviour, and socio-demographic characteristics. [fn: Details of the BES 2019 methodology, questionnaire, and data access are available at www.britishelectionstudy.com. The wave used here is the post-election face-to-face survey.] It contains a large number of orcats — items measured on ordered scales with between 4 and 11 categories — that are routinely correlated and factor-analysed in published research. We identified all orcats in the BES 2019 with between 4 and 11 categories and at least minimal variance, yielding a total of 7,503 variable pairs. For each pair we computed $r_{\min}$, $r_{\max}$, C_1, C_2, and the asymmetry measure $|r_{\min}| - r_{\max}$ from the observed marginal frequencies using the comonotonic and anti-comonotonic constructions of Section 3. We do not claim that the BES is a random sample of all survey datasets used in social science, but we do regard it as typical: it is a large, methodologically sophisticated survey containing the variety of item formats, response distributions, and category counts that applied researchers encounter routinely.

6.1 The Attainable Bounds Across All Variable Pairs

Figure 4 displays the full joint distribution of $r_{\min}$ and $r_{\max}$ across all 7,503 BES pairs. Each point is one variable pair; the dashed line marks $r_{\max} = -r_{\min}$ (i.e., perfect symmetry of bounds).

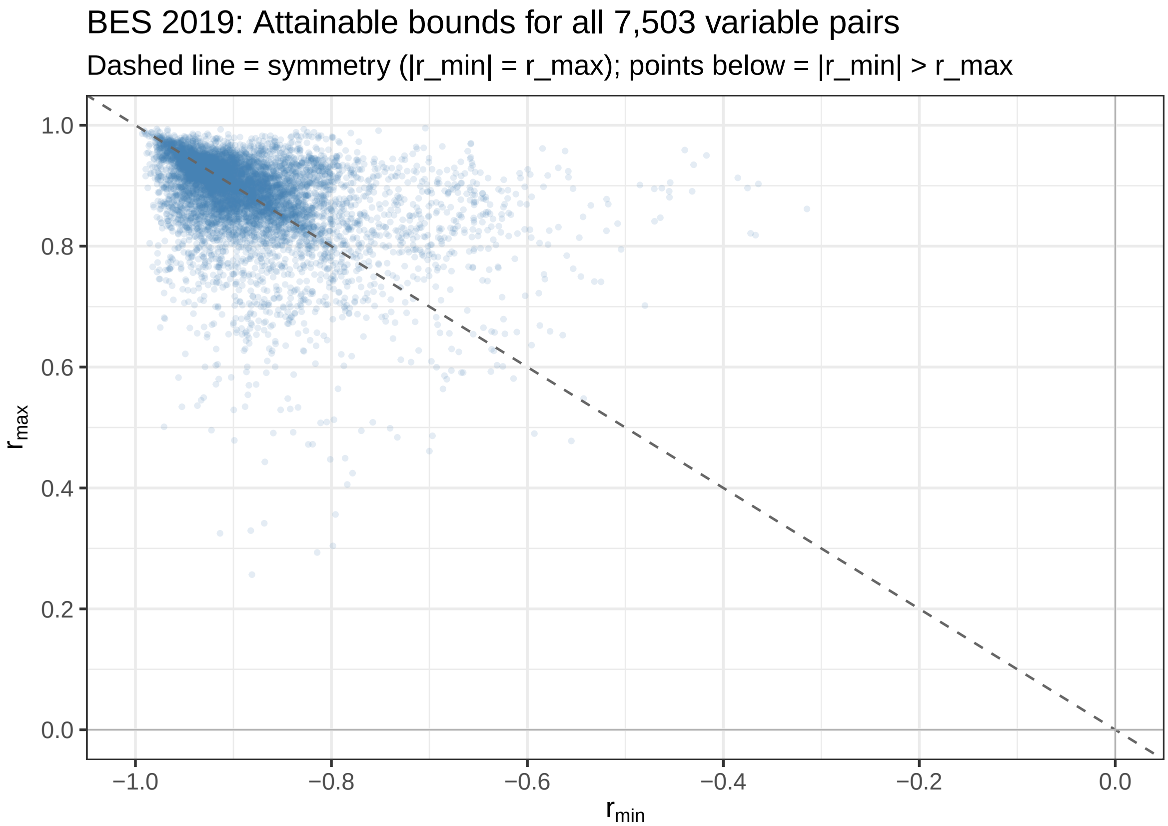

Figure 4: Attainable bounds $(r_{\text{min}}, r_{\text{max}})$ for all 7,503 variable pairs in the BES 2019. Each point is one pair. The dashed diagonal marks perfect symmetry of bounds ($r_{\text{max}} = |r_{\text{min}}|$); points above this line have $r_{\text{max}} > |r_{\text{min}}|$ and points below have $|r_{\text{min}}| > r_{\text{max}}$. Both axes are restricted to their feasible ranges: $r_{\text{min}} \in [-1, 0]$, $r_{\text{max}} \in [0, 1]$.

Three features of this figure deserve attention.

First, no pair in the BES has bounds reaching $[-1, +1]$. The most tightly constrained pairs have $r_{\text{max}} \approx 0.26$ and $r_{\text{min}} \approx -0.32$, representing an attainable range roughly one-third the width of $[-1, +1]$. The least constrained pairs reach close to but do not attain $r_{\text{max}} = 1$ or $r_{\text{min}} = -1$. Every single pair in a large and representative survey dataset operates on a different, sub-maximal reference interval.

Second, the bulk of the cloud lies in the upper-left region — high r_{max} (around 0.85–0.99) paired with highly negative r_{min} (around -0.85 to -0.99). This reflects the fact that BES items tend to have moderate to high variance (survey designers avoid degenerate items), which in turn produces marginals with substantial entropy and thus less-constrained bounds. The mean C_1 in the BES is 0.885, meaning that the typical variable pair covers about 88.5% of the theoretical $[-1, +1]$ range.

Third, there is a pronounced horizontal spur extending from the main cloud toward the right: a set of pairs where r_{max} remains high (around 0.85–0.95) but r_{min} is considerably less negative, ranging from -0.7 down to -0.3. Examining these 205 spur pairs (2.7% of all BES pairs) reveals a distinctive profile: both variables have strong positive skewness (median skewness 1.13 for the first variable, 0.92 for the second) and both have high total variation distance between their marginal distribution and its reversal (median TV 0.76 and 0.66 respectively), compared to the main cloud (median TV 0.39 and 0.34, median skewness near zero). The mechanism follows directly from the theoretical framework. When both marginals concentrate mass at the low end of the scale, the comonotonic coupling pairs these abundant low values together, achieving a strong positive correlation and a high $r_{\max}$. The anti-comonotonic coupling, however, must pair the sparse high-value observations of X against those of Y; since both distributions are thin at the upper end, the absolute negative covariance this pairing can generate is limited, resulting in a $r_{\min}$ that is much less negative than $r_{\max}$ is positive. The spur is therefore a signature of joint positive skewness: both marginals skewed in the same direction simultaneously inflates $r_{\max}$ relative to $|r_{\min}|$. Consistent with this, all 205 spur pairs satisfy $r_{\max} > |r_{\min}|$ — the asymmetry is exclusively in one direction — whereas among main-cloud pairs the direction split is nearly 50/50.

6.2 Bound Asymmetry in the BES

Figure 5 shows the distribution of bound asymmetry, $|r_{\min}| - r_{\max}$, across all BES pairs.

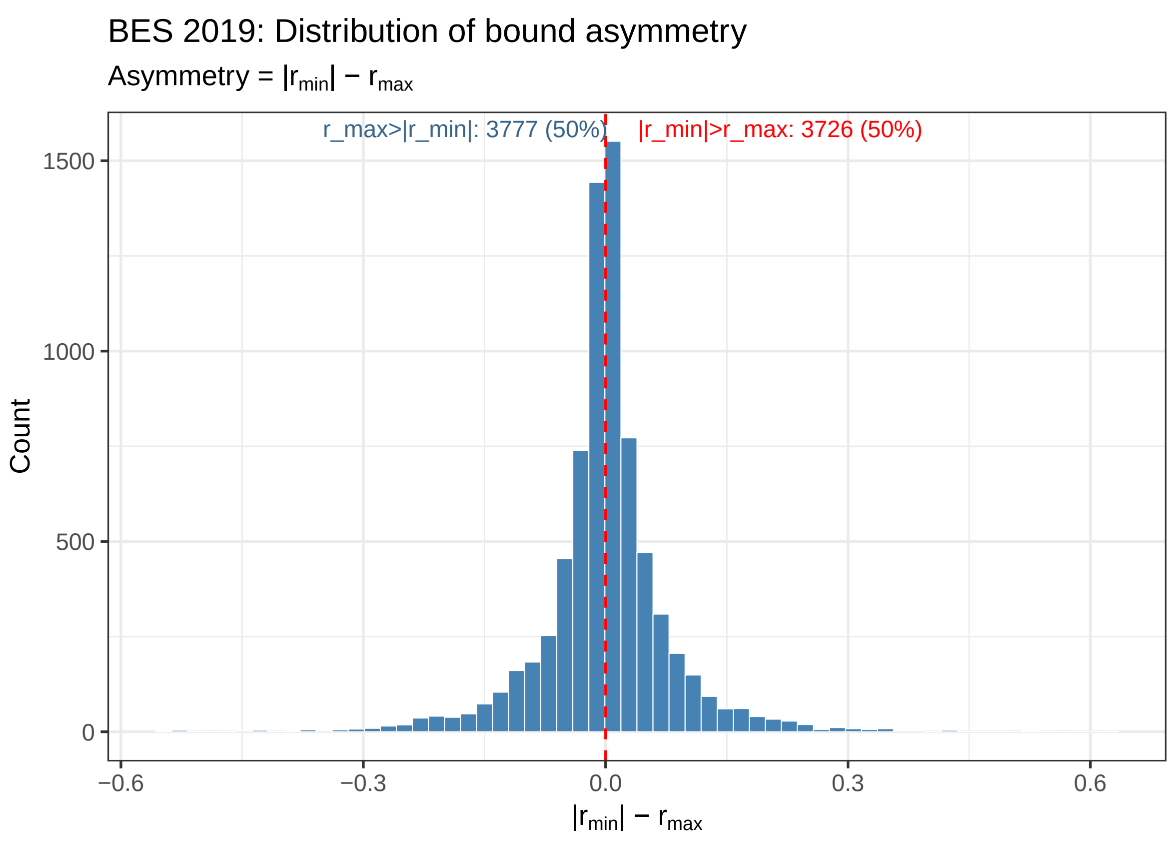

Figure 5: Distribution of bound asymmetry, $|r_{\min}| - r_{\max}$, across 7,503 BES 2019 variable pairs. Positive values indicate $|r_{\min}| > r_{\max}$; negative values indicate $r_{\max} > |r_{\min}|$. The vertical red dashed line marks exact symmetry at zero.

The distribution is nearly symmetric about zero: 3,726 pairs (49.7%) have $|r_{\min}| > r_{\max}$ and 3,777 pairs (50.3%) have $r_{\max} > |r_{\min}|$. The split is essentially a coin flip, confirming the simulation finding of Section 5: the *direction* of bound asymmetry cannot be predicted from univariate marginal summaries.

The *magnitude* of asymmetry is more informative. The distribution has a sharp peak at zero — most pairs are nearly symmetric — but the tails extend to ± 0.4, meaning that for some pairs the difference between $|r_{\min}|$ and $r_{\max}$ is as large as 0.4 correlation units. For such pairs, an analysis that assumes symmetric bounds (i.e., treats $r_{\min} = -r_{\max}$) would be wrong by a large margin. The BES example introduced in Section 3 — where $r_{\max} = 0.850$ and $r_{\min} = -0.899$, an asymmetry of 0.049 — is representative of the more common modest asymmetries, while the tails of Figure 5 contain pairs where the issue is substantially more severe.

Since asymmetry cuts both ways with equal frequency, one cannot presume that the standard assumption $r_{\min} \approx -r_{\max}$ introduces a systematic bias in any particular direction. Rather, it introduces *noise*: for any given pair, the applied researcher who does not compute the bounds cannot know whether the true attainable interval is $[-0.85, 0.85]$, $[-0.90, 0.85]$, or $[-0.80, 0.85]$ — and the difference matters for both effect-size interpretation and hypothesis testing.

6.3 Comparing BES Constraint Severity to the Simulation

Figure 6 overlays the C_1 distributions from the MC simulation and the BES, allowing a direct comparison between the full theoretical space of orcat pairs and those encountered in an actual survey.

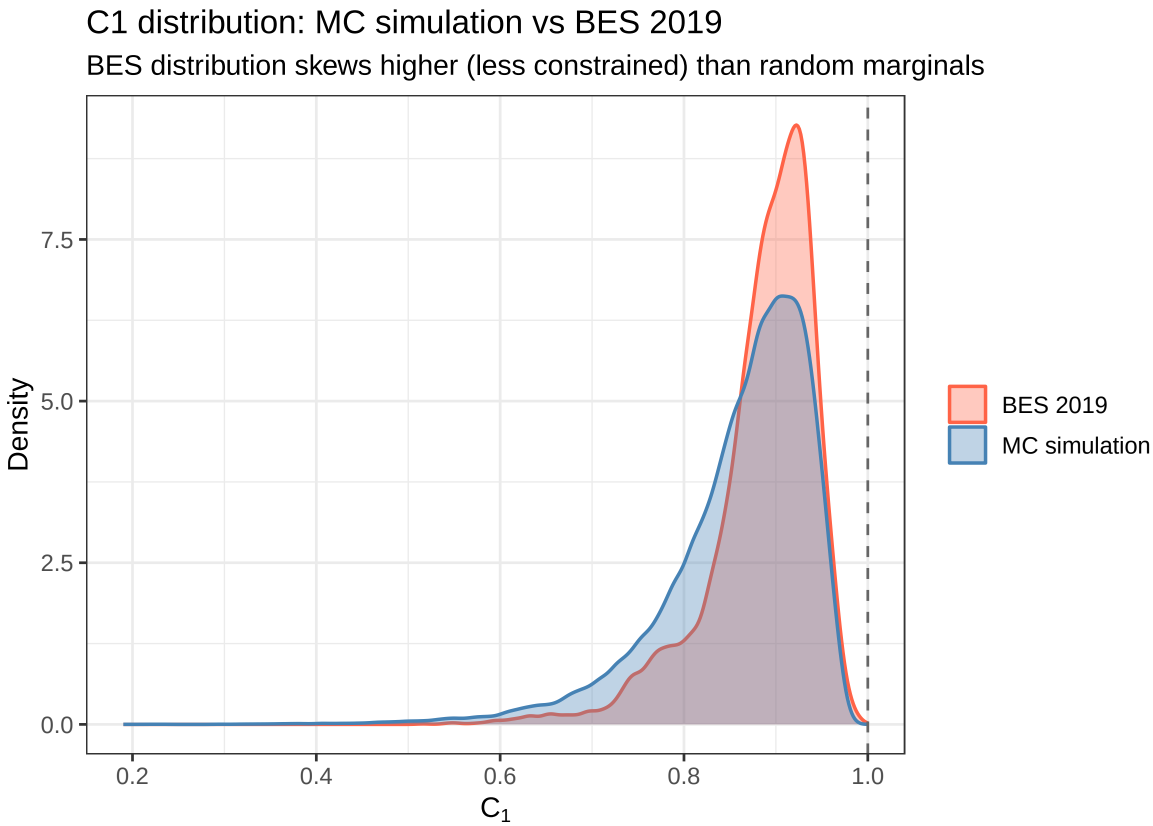

Figure 6: Distribution of constraint severity (C_1) in the BES 2019 (red/orange) and in the MC simulation (blue). The simulation draws from all possible marginal shapes for $K \in \{4, 5, 6, 7, 10, 11\}$; the BES contains the marginal shapes actually observed in a large-scale political survey. The dashed vertical line at $C_1 = 1$ marks the unconstrained reference.

Two findings stand out.

First, the BES distribution is shifted right relative to the simulation, with its mode around $C_1 = 0.91$ compared to $C_1 \approx 0.88$ in the simulation. This is expected: the simulation draws from the full Dirichlet simplex, including highly concentrated marginals that survey researchers would avoid, while the BES reflects the distribution of marginals that actually arise when questions are carefully designed to elicit variance. Applied researchers are therefore less likely to encounter the most severe constraints shown in the left tail of the simulation distribution, but those constraints are not absent from the BES either — the lower tail of the BES C_1 distribution extends below 0.55.

Second, despite the offset, both distributions have essentially the same shape and span a similar range. This suggests that the simulation provides a reasonable model of the space of constraints that applied researchers face, with the important caveat that very severe constraint ($C_1 < 0.6$) is somewhat more common in the theoretical space than in practice. For the purposes of understanding what typically drives constraint in surveys, the BES and the simulation tell a consistent story: constraint is pervasive, its severity varies considerably, and it is primarily driven by the number of categories and the entropy of the marginal distributions.

7 Implications for Applied Research

The theoretical and empirical results of the preceding sections establish that the attainable range of Pearson's r for ordinal-categorical variable pairs depends on marginal distributions in ways that are substantial, variable, and not easily predicted from simple summaries of the data. These findings carry direct consequences for how correlations are interpreted and used in applied research. This section develops those consequences across five domains: effect size interpretation, comparability across studies, hypothesis testing, rescaling, and variance explained. Each implication points toward the same underlying problem — that treating r as though it operates on the universal $[-1, +1]$ scale introduces systematic distortions that can be diagnosed and, in many cases, corrected.

7.1 Interpretation of Effect Size

The benchmarks for correlation effect size proposed by Cohen et al. (2002) — small ($r \approx 0.10$), moderate ($r \approx 0.30$), and large ($r \approx 0.50$) — have become standard reference points in the social sciences. Their implicit premise, however, is that the correlation coefficient operates on its full theoretical range $[-1, +1]$: a "large" effect of $r = 0.50$ is large precisely because it represents half of the maximum achievable association. When the attainable range for a given variable pair is not $[-1, +1]$ but the pair-specific interval $[r_{\min}, r_{\max}]$, this premise fails. An observed correlation must be evaluated against the pair's actual attainable range.

The risk of understating association is most acute for highly constrained pairs. Consider a pair with $r_{\min} = -0.55$ and $r_{\max} = 0.62$ ($C_1 = 0.585$, well within the range observed in the BES). An observed $r = 0.45$ lies at $0.45/0.62 \approx 73\%$ of the maximum attainable — a strong effect given the constraints, yet below Cohen's "large" threshold. A naive application of the benchmarks classifies it as moderate. Conversely, a pair with $r_{\max} = 0.98$ and the same observed $r = 0.45$ is a genuinely moderate effect at 46% of its maximum. The same numerical value of r carries fundamentally different substantive meaning depending on the attainable range.

The BES 2019 data illustrate the practical scope of this problem. Across 7,503 variable pairs, C_1 ranges from 0.52 to 0.99 with a median of 0.885. Even at the median, the effective "large" threshold shifts: a pair with $r_{\max} = 0.88$ requires an observed $r \approx 0.44$ to achieve the same proportional position as Cohen's 0.50 in the unconstrained case. For the most constrained pairs — those in the lower quartile of C_1 — the distortion is far more severe. The rescaled correlation r^* (Section 7.4) resolves this directly: by expressing r_{obs} as a proportion of the pair-specific attainable range, Cohen-style benchmarks applied to r^* recover their intended meaning.

7.2 Comparison Across Studies

Non-comparability also arises from differences in marginal distributions, even when category counts are held fixed. Survey items that appear identical may elicit different response distributions across populations, time points, or question wordings. Left-right self-placement, for instance, is characteristically more peaked in some countries than others; the resulting difference in marginal distributions shifts both r_{min} and r_{max}, altering the attainable range even when the substantive association is unchanged. Two studies reporting $r = 0.35$ for the same pair of attitude items may be observing very different degrees of association once pair-specific bounds are computed.

The same logic applies to coding decisions within a single study. Collapsing a 7-point scale to 5 points changes the marginal distribution and therefore changes r_{min}, r_{max}, and C_1, even if the underlying association is unchanged. Researchers who compare correlations before and after such recoding are comparing values on incommensurable scales. Bound-aware comparison — reporting r alongside $(r_{\text{min}}, r_{\text{max}})$, or reporting r^* directly — provides the common reference frame that raw correlations alone cannot supply.

7.3 Hypothesis Testing

The standard test of $H_0\colon \rho = 0$ for Pearson's r relies on the t-statistic $t = r\sqrt{n-2}/\sqrt{1-r^2}$, which under the null follows a t-distribution with $n-2$ degrees of freedom when the variables are jointly normal and, by continuous-data theory, r is free to range over $(-1, 1)$. For two orcat variables the second condition fails: the marginal distributions place an iron constraint on the set of correlations that can arise in any arrangement of the data, confining r to the interval $[r_{\text{min}}, r_{\text{max}}]$. Applying the conventional t-test therefore attributes significance to variation that is, in part, the inevitable consequence of the marginal distributions rather than of any genuine dependence in the joint distribution.

A natural alternative is a permutation (randomisation) test that respects the marginal constraint. Under $H_0\colon \rho = 0$ the appropriate reference distribution is obtained by independently permuting one of the two variables while holding the marginals fixed, computing r for each permuted data set, and repeating many times. The resulting permutation null distribution is centred near zero — its exact centre depends weakly on the category scores — and is entirely contained within $[r_{\text{min}}, r_{\text{max}}]$. A p-value is then computed as the proportion of permuted r-values that are at least as extreme as the observed r_{obs} in absolute value. Because the reference distribution respects the constraints, the permutation test is valid unconditionally on the marginal distributions.

The key practical insight is not merely that the permutation p-value can differ from the t-test p-value — though it can — but that the permutation distribution renders transparent where r_{obs} falls within the pair-specific attainable range. Two variable pairs that share the same r_{obs} and the same t-test p-value can occupy very different positions within their respective $[r_{\text{min}}, r_{\text{max}}]$ intervals, and this positional information is invisible to the conventional test.

Figure 7 illustrates the contrast with two synthetic variable pairs of identical sample size ($n = 1{,}000$), both yielding statistically significant correlations by the standard t-test. Pair~A consists of two 4-category orcats with heavily concentrated, opposing marginals ($r_{\min} = -1.00$, $r_{\max} = 0.33$, $C_1 = 0.66$). The observed correlation is $r_{\text{obs}} = 0.17$, which equals 52% of the pair's theoretical maximum: the association accounts for a substantial fraction of what the marginals allow. Pair~B consists of two 7-category orcats with near-uniform marginals ($r_{\min} = -0.99$, $r_{\max} = 1.00$, $C_1 = 0.99$). Its observed correlation is $r_{\text{obs}} = 0.13$, which is only 12.7% of the theoretical maximum: the association is barely above what would arise by chance given the marginals. Despite their superficial similarity as "significant correlations around 0.15", the two pairs convey qualitatively different information — a distinction that the permutation null distributions make immediately visible.

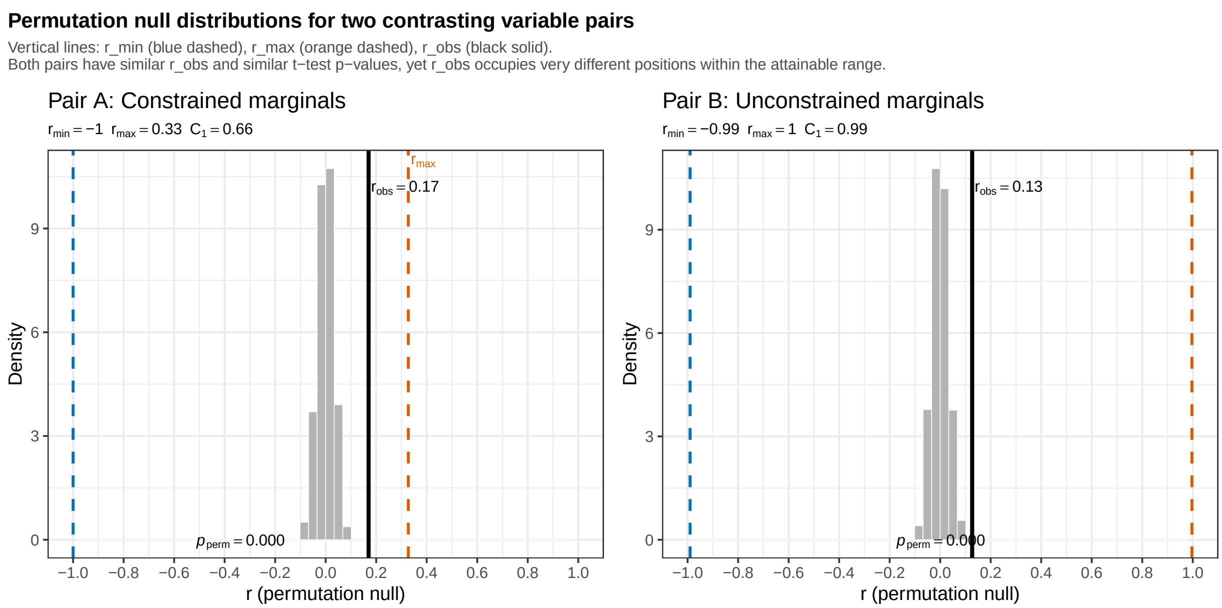

Figure 7: Permutation null distributions for two contrasting variable pairs, both with $n = 1{,}000$ and similar r_{obs} values. Pair~A (left) has concentrated, opposing marginals ($C_1 = 0.66$); Pair~B (right) has near-uniform marginals ($C_1 = 0.99$). Vertical lines mark $r_{\min}$ (blue dashed), $r_{\max}$ (orange dashed), and r_{obs} (black solid). Although both pairs yield a nominally significant t-test, r_{obs} occupies 52% of the attainable range for Pair~A but only 13% for Pair~B.

A further subtlety concerns the *shape* of the permutation null distribution under asymmetric bounds. When $|r_{\min}| \neq r_{\max}$ the distribution is not symmetric around zero, and using a two-sided critical region of equal tail probability on $[-1, 1]$ — as the t-test implicitly does — can produce distorted type~I error rates. The permutation test is immune to this distortion because its critical region is derived directly from the empirical null distribution, whatever shape that takes.

In summary, permutation-based inference is the natural complement to bound-awareness in the exploratory phase: it does not require any distributional assumption beyond the fixed marginals, it automatically adapts to the pair-specific attainable range, and it makes visible the position of r_{obs} within that range — information that is entirely suppressed by the conventional t-test.

7.4 Rescaling

The most direct response to the comparability problem is to rescale each observed correlation into the $[-1, +1]$ interval by expressing it as a proportion of its attainable range. A natural linear rescaling that maps r_{min} to -1, r_{max} to $+1$, and preserves the sign of zero is:

$$r^* = \frac{r_{\text{obs}} - r_{\text{min}}}{r_{\text{max}} - r_{\text{min}}} \times 2 - 1.$$

This transformation places every variable pair on the same $[-1, +1]$ reference interval, making correlations from pairs with different numbers of categories and different marginal distributions directly comparable.

Figure 8 shows the relationship between r_{obs} and r^* across all 7,503 BES pairs.

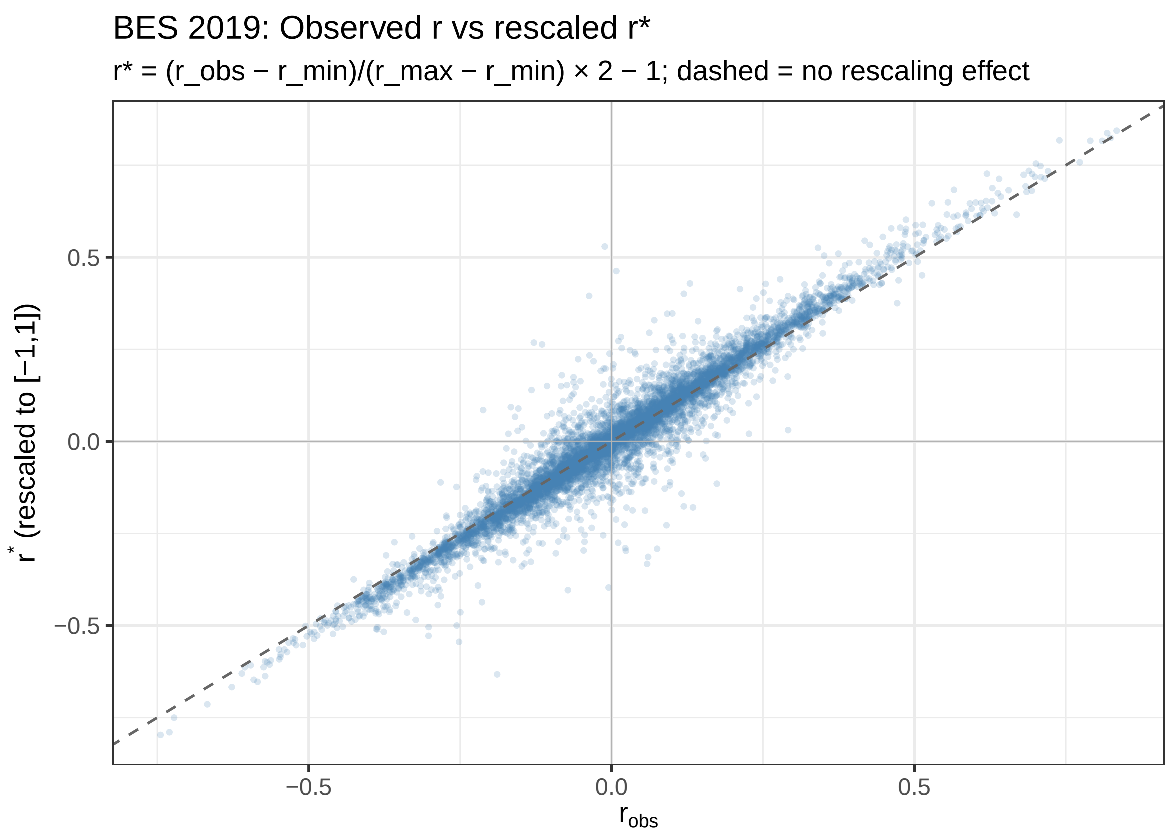

Figure 8: Observed Pearson correlation r_{obs} versus bound-rescaled correlation r^* for all 7,503 BES 2019 variable pairs. The dashed diagonal marks no rescaling effect ($r^* = r_{\text{obs}}$). Points above the diagonal have been amplified by rescaling; points below have been attenuated.

Two findings from Figure 8 are important for applied practice.

First, for most BES pairs the rescaling adjustment is modest: the cloud of points lies close to the diagonal, reflecting the fact that the typical BES pair has $C_1 \approx 0.885$ and relatively symmetric bounds, so the transformation compresses or expands the scale only slightly. This should not be taken as evidence that computing bounds is unnecessary — the *direction* of adjustment varies across pairs, and for the pairs with $C_1 < 0.7$ or large bound asymmetry the adjustment can be substantial (exceeding 0.15 in absolute terms for some pairs).

Second, and more fundamentally, the *rank ordering* of correlations changes after rescaling. Two pairs with the same observed r can have very different r^* values if their attainable ranges differ. Consider a pair (A, B) with $r_{\text{obs}} = 0.35$ and $r_{\text{max}}^{AB} = 0.45$ (near its theoretical ceiling) versus a pair (C, D) with $r_{\text{obs}} = 0.35$ and $r_{\text{max}}^{CD} = 0.85$ (far from its ceiling): after rescaling, $r^*_{AB} \approx 0.78$ while $r^*_{CD} \approx 0.12$. The first pair is effectively a strong association given the constraints; the second is a weak one. This reversal — visible in the scatter of Figure 8 around the diagonal — is the core practical problem that examining bounds addresses.

Rescaling to r^* does not resolve every difficulty. It assumes that the attainable range $[r_{\text{min}}, r_{\text{max}}]$ can be treated as the appropriate reference interval for the specific pair, which is correct but does not address the fact that r^* itself still depends on the arbitrary choice of consecutive-integer scoring for the categories (see Section 2). It also does not address the asymmetry problem directly: when $|r_{\text{min}}| \neq r_{\text{max}}$, the rescaled zero is not the same as $r_{\text{obs}} = 0$. Adjusted null distributions via permutation testing (Section 7.3) provide a complementary approach that avoids these artefacts.

7.5 Variance Explained

- Interpretation of r^2 is ambiguous for at least two reasons:

1. ordcats are invariant to monotone transformation (variance is a *cardinal* property, simply not appropriate for ordinal variables)
2. is r^2_{max} the .... maximal 'amount' of variance between the two ordcats? How is this quantity to be interpreted? How is r^2_{min} to be interpreted?

7.6 Correlation Matrices, PCA, Factor Analysis, and SEM

The consequences of using Pearson correlations with ordinal data extend beyond descriptive association. In factor analysis and SEM, attenuated correlations can yield distorted loadings, inflated factor numbers, and biased structural coefficients. Empirical studies confirm this risk: (Flora & Curran, 2004) show that CFA models fitted with Pearson's r among ordinal indicators produce biased estimates compared to models using polychoric correlations; (Green et al., 1997) demonstrate how the number of response categories alters apparent model fit. More broadly, psychometric research emphasizes that ignoring the ordinal nature of indicators can misrepresent latent structures (Bartholomew et al., 2011; Lord, 1952; Muthén, 1984). For survey-based political science and sociology, this means that correlations reported between Likert items are not only bounded by their marginals, but may also systematically bias downstream modeling.

Since the possible values r may take on depend on the marginal distributions of the two variables being correlated, and since this reference interval may – and often is – quite different for different pairs of variables, interpreting and working with correlation matrices (alternatively variance-covariance matrices) must be paid special attention. Such matrices form the basis for a number of statistical techniques, from regression to Factor Analysis or Structural Equation Modeling (CITES). In this section we present preliminary results exploring the implications of our findings above for PCA, FA, SEM.

First, some preliminaries. It is well known that any correlation matrix (using complete cases – i.e. when each pair of variables is observed for the same set of respondents), S, is positive semi-definite and symmetric. Hence, S is invertible. However, …

Why PSD Matters for PCA:

- **Mathematically**: Eigen-decomposition works for any symmetric $\mathbf{S}$, PSD or not.
- **Interpretively**: Negative eigenvalues break the "variance explained" story (variance cannot be negative).
- **Software behavior**: Most PCA routines check PD/PSD; failure can cause warnings or dropped components.

If Input is Symmetric but not PSD:

Example:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \quad \lambda = (3, -1),$$

is indefinite (one negative eigenvalue).

- PCA math still runs; eigenvalues/vectors computed.
- Interpretation fails: $\sqrt{\lambda_j}$ imaginary for $\lambda_j < 0$.
- Some software refuses input; others run but return nonsensical variance metrics.

PSD/PD Needs in FA:

- **ML FA**: Requires $\mathbf{S}$ be PSD; $\log\det\mathbf{S}$ undefined if indefinite.
- **Principal-axis / Image FA**: Also break with negative eigenvalues (square roots fail).

SEM and PSD Requirements

7.6.1 SEM Machinery

Observed covariance $\mathbf{S}$; implied covariance $\Sigma(\theta)$ from structural and measurement models.

7.6.2 Fit Functions and PD Needs

- **ML / GLS**: Require both $\mathbf{S}$ and $\Sigma(\theta)$ PD.
- **WLS/DWLS**: Need PD indirectly (weight matrix invertible).
- **ULS**: Can start from indefinite $\mathbf{S}$, but test statistics again require PD.

7.6.3 What Happens if Not PSD

- Software errors out ("matrix is not positive definite").
- Log-determinant undefined; inversion fails.
- Even if bypassed, optimisation steps hit complex/NaN values.

7.6.4 Remedies in SEM

- Nearest PD smoothing (Higham 2002).
- Ridge adjustment.
- Analyse raw data.
- Check rounding issues.

Core Lessons Across PCA, FA, SEM

1. **Theoretical PSD**: Covariance matrices from real data are PSD by definition.
2. **Numerical Issues**: Sampling noise, rounding, or listwise deletion can yield small negative eigenvalues.
3. **PCA**: Can compute from indefinite matrix but interpretation breaks.
4. **FA and SEM**: Require PD input; will not run on indefinite matrices.
5. **Repairs**: Adjust to nearest PSD, ridge, or recompute from raw data.

So, we consider linearly re-scaling the estimated r for each pair of variables in the correlation matrix S. Call this new rescaled correlation matrix S'. We seek to understand what the process of pair-wise linear rescaling has on the PSD-ness and invertibility of S'. This is important because without an invertible correlation matrix, several analyses scholars perform on S become questionable (at best). Further, some estimators used in e.g. FA require PSD (namely, MLE).

The general result is that linear rescaling does not 'break' invertibility (nor PSD) but does make invertibility 'worse' vis-à-vis the condition number. [Discuss condition number here]

8 Discussion

8.1 Comparing 'Typical' Data (BES) with Simulated Data

8.2 Relation to Literature

!fullcite camb76-inequalities

Full citation:

Cambanis et al. (1976)

Grether, D. M. (1976). On the Use of Ordinal Data in Correlation Analysis. American Sociological Review, 41(5), 908–912. https://doi.org/10.2307/2094741

Labovitz, S. (1967). Some observations on measurement and statistics. Social Forces, 46(2), 151–160. https://doi.org/10.2307/2574595

Labovitz, S. (1970). The assignment of numbers to rank order categories. American Sociological Review, 35(3), 515–524. https://doi.org/10.2307/2092993

Moss, J., & Grønneberg, S. (2023). Partial Identification of Latent Correlations with Ordinal Data. Psychometrika, 88(1), 241–252. https://doi.org/10.1007/s11336-022-09898-y

O'Brien, R. M. (1979). The use of Pearson's r with ordinal data. American Sociological Review, 44(5), 851–857. https://doi.org/10.2307/2094532

Robitzsch, A. (2022). On the Bias in Confirmatory Factor Analysis When Treating Discrete Variables as Ordinal Instead of Continuous. Axioms, 11(4), Article 4. https://doi.org/10.3390/axioms11040162

Fréchet (1951)

Whitt (1976)

Hoeffding (1940)

Lehmann (1966)

Tchen (1980)

Cambanis et al. (1976)

Shih and Huang (1992)

Alwin and Krosnick (1991); Brewer and Gross (2005); Flora and Curran (2004); Gaito (1980); Green et al. (1997); Jacoby (1999); Kendall (1938); Knoke et al. (2002); Kreuter and Valliant (2007); Krosnick and Fabrigar (1997); Labovitz (1970); Olsson (1979); Spearman (1904)

8.3 Recommendations

- Bound-aware effect size interpretation.
- Bound-aware hypothesis testing.
- Possible inclusion in statistical software.

9 Conclusion

Summarize:

- Main theoretical findings.
- Key applied consequences.
- Summary: Theoretical bounds, symmetry-breaking conditions, and implications for applied research
- An R package accompanies this paper, providing tools to compute max/min bounds and perform randomization-based hypothesis tests for fixed marginals. Details are available in the appendix and online.

Impacts standard methods:

- interpretation (e.g. ‘effect size’) and comparison
- hyothesis testing

Recommendations for interpreting Pearson’s r in applied contexts

- Reporting observed r in context of its attainable bounds.
- Using our tools to test whether an observed r is unusually strong or weak given marginals.

9.1 Extensions and Open Questions

- Role of entropy and joint uncertainty in bounding behavior

- Possibility of adjusting or normalizing r
- Relation to Copula Theory (Optional)
  - “Although our setting is discrete, the problem is conceptually linked to copula-based modeling of dependence with fixed marginals. The connection is discussed as a direction for future work.”
- ‘Correction’ strategies $\rightsquigarrow$ *rescaling*
- Generalizations to higher dimensions $\rightsquigarrow$ see Section 7.6 above for implications for correlation matrices, FA, and SEM

10 References

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A Appendix A: Mathematical Proofs

A.1 Proof A.1

Proof of prop-max-product<empty citation>.

[Proof to be written. The argument is given in full in the body of Section 3; this appendix will present a self-contained, condensed version.]

A.2 Proof A.2

Proof of the Corollary (corr-min<empty citation>).

[Proof to be written. The greedy algorithm for constructing $M_{i,j}$ and $R_{i,j}$ is described in the body of Section 3; this appendix will provide a formal verification that the construction satisfies the marginal constraints and achieves the claimed extremal expectations.]

B Appendix B: Software Tools and Code Listings

- Documentation and examples for the R package
- Description of randomization testing features

C Appendix C: Additional Tables and Simulations

- Supplementary empirical examples, simulation results